

Hochiminh city University of Technology
Faculty of Computer Science and Engineering



COMPUTER GRAPHICS

CHAPTER 01:

Graphics System

Outline

- ❑ Computer Graphics: Why & What?
- ❑ Application
- ❑ Computer Graphics Systems
- ❑ Image Formation
- ❑ Models and Architectures
- ❑ API Contents

Computer Graphics: Why & What?

□ Why

– “A Picture is Worth a Thousand Words”



How does a water softener work?
by Nicholas Gerbis

Share Tweet 83 +1 33 Page 1 2 3

howstuffworks.com
Brain Stuff
with Marshall Brain
Water Softner

Watch as Marshall Brain explains how water softeners work.
brainstuff

UP NEXT

- How Home Dry Cleaning Works
- How the Toto Washlet Works
- 10 Stand-up Facts About Waterless Urinals

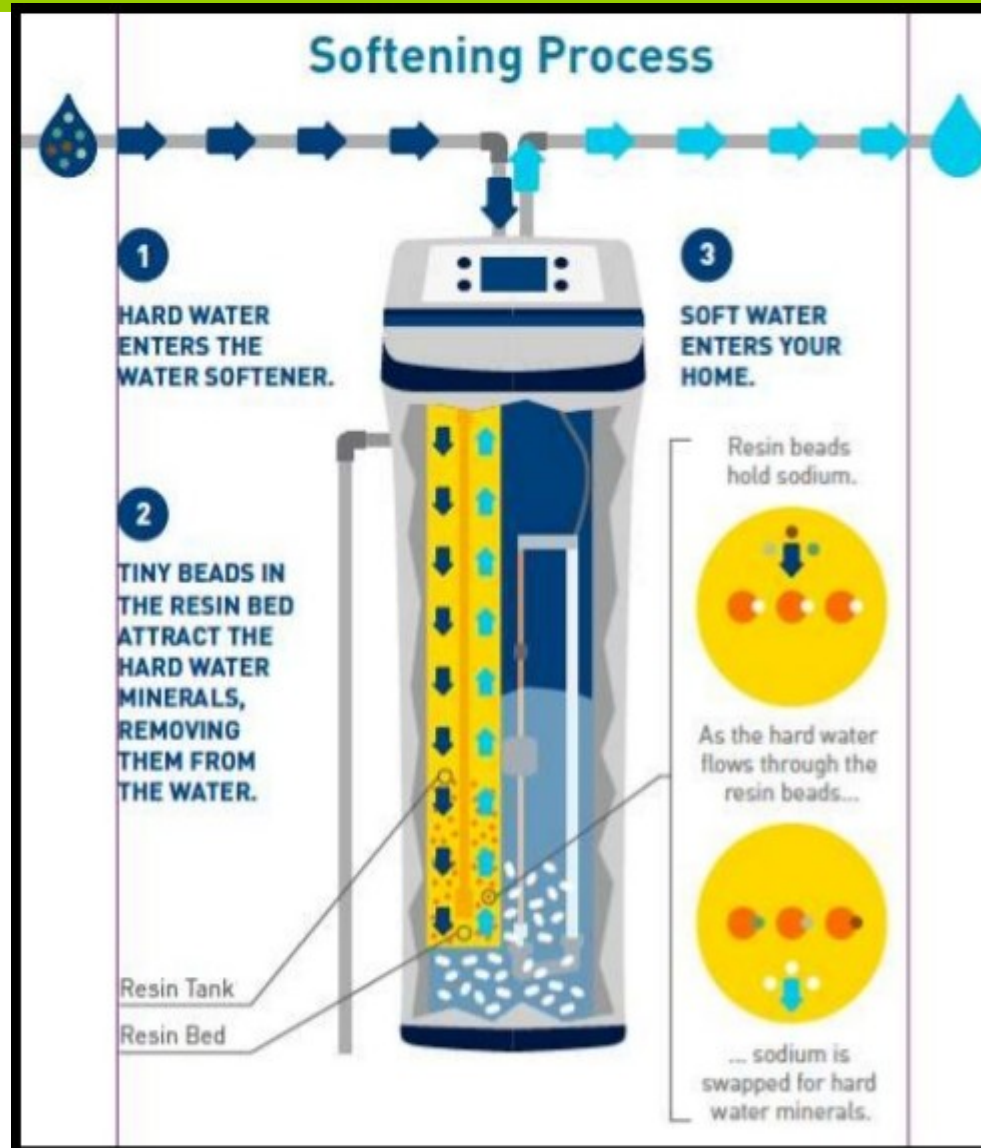
Quick: What's hard and scaly and dwells in your pipes? No, it's not the pet alligator your parents flushed down the toilet -- he's grown up and terrorizing the sewers of Chicago. We're talking about hard water.

We call water "hard" if it contains a lot of calcium, magnesium or other minerals. Groundwater acquires these metals by dissolving them from surrounding soil and rock. Industry measures water hardness in terms of grains per gallon (GPG) or milligrams per liter (mg/L). A grain is defined as 64.8 milligrams of calcium carbonate [source: [Business Dictionary](#)]. If your water tests at 1 GPG (17.1 mg/L) or less, then you have soft water. Water around 1-3.5 GPG (17.1-60 mg/L) occupies a gray zone between soft and slightly hard water and 3.5-7 GPG (60-120 mg/L) is moderately hard. Hard water is around 7-10.5 GPG (120 - 180 mg/L), and very hard water is above that [source: [Water Quality Association](#)].

How do all those number affect you? Hard water causes two problems:

1. Dissolved calcium and magnesium precipitate out of hard water as **scale**, which builds up on the insides of pipes, **water heaters**, **tea** kettles, coffee makers and industrial machinery. Scale reduces flow through pipes and is a poor conductor of heat. Eventually, pipes can become completely clogged.
2. Hard water reduces soap's ability to lather, whether in the shower, sink, dishwasher or washing machine, and reacts with soap to form a sticky scum.

Computer Graphics: Why & What?



Computer Graphics: Why & What?



Computer Graphics: Why & What?



**THE HUMAN BRAIN PROCESSES VISUALS
60,000 TIMES FASTER THAN TEXT.**

Computer Graphics: Why & What?

□ What

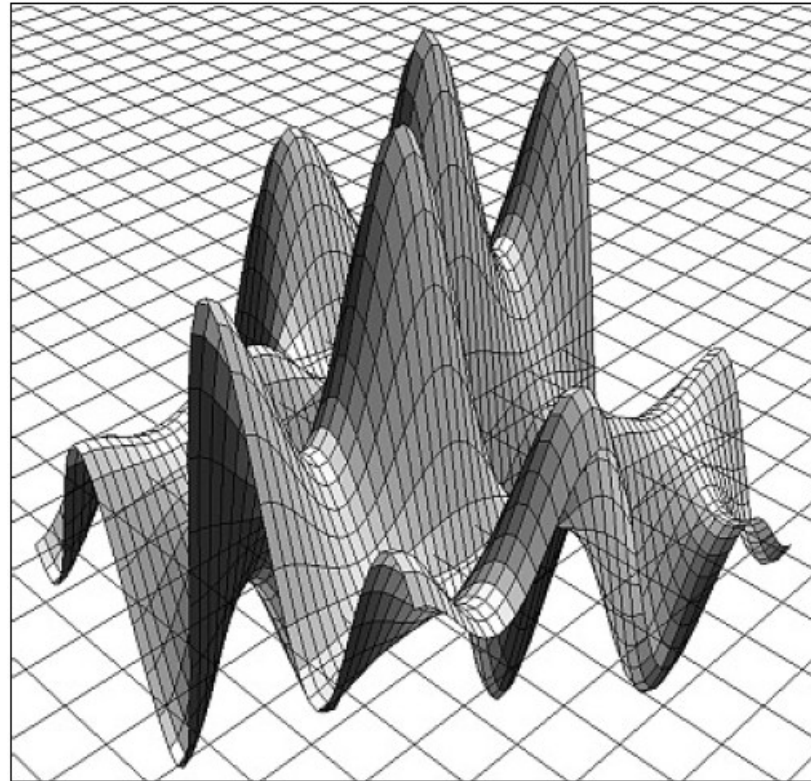
- *Computer graphics* deals with all aspects of creating images with a computer
- **Hardware**
- **Software**
 - High-Level: Maya, Lightwave, 3DS Studio
 - Low-level: OpenGL, Direct3D – Libraries for programming graphics applications
- **Applications**

Application

- ☐ Display of Information
- ☐ Computer-Aided Design
- ☐ Simulation and Animation
- ☐ User Interface

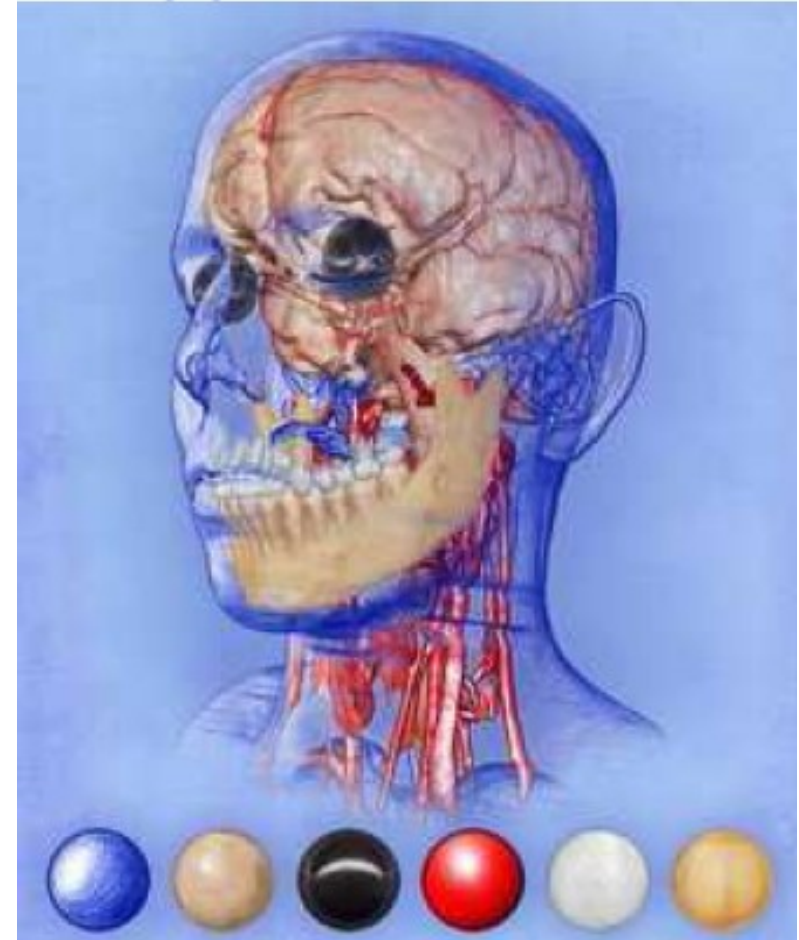
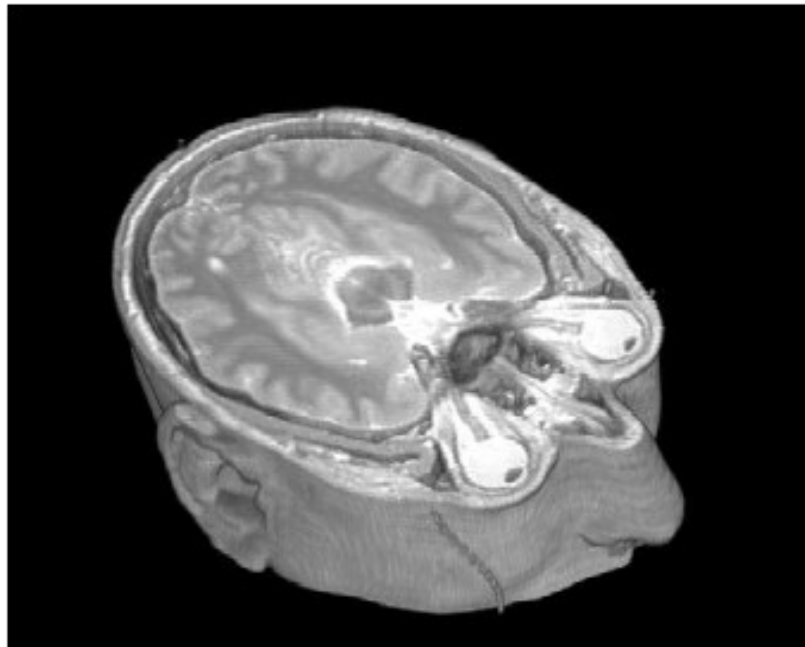
Application

- ❑ Display of Information
 - Complex scientific data



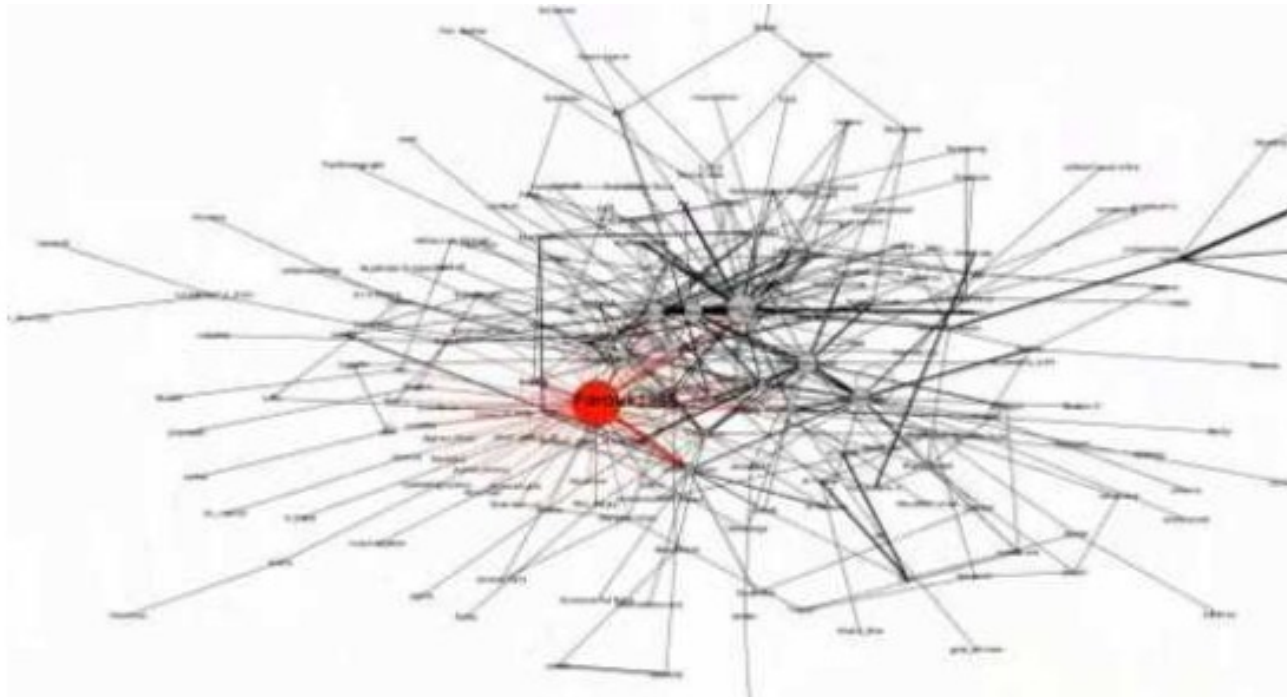
Application

- ❑ Display of Information
 - **Medical Imaging**



Application

- ❑ Display of Information
 - **Network and threat visualization**



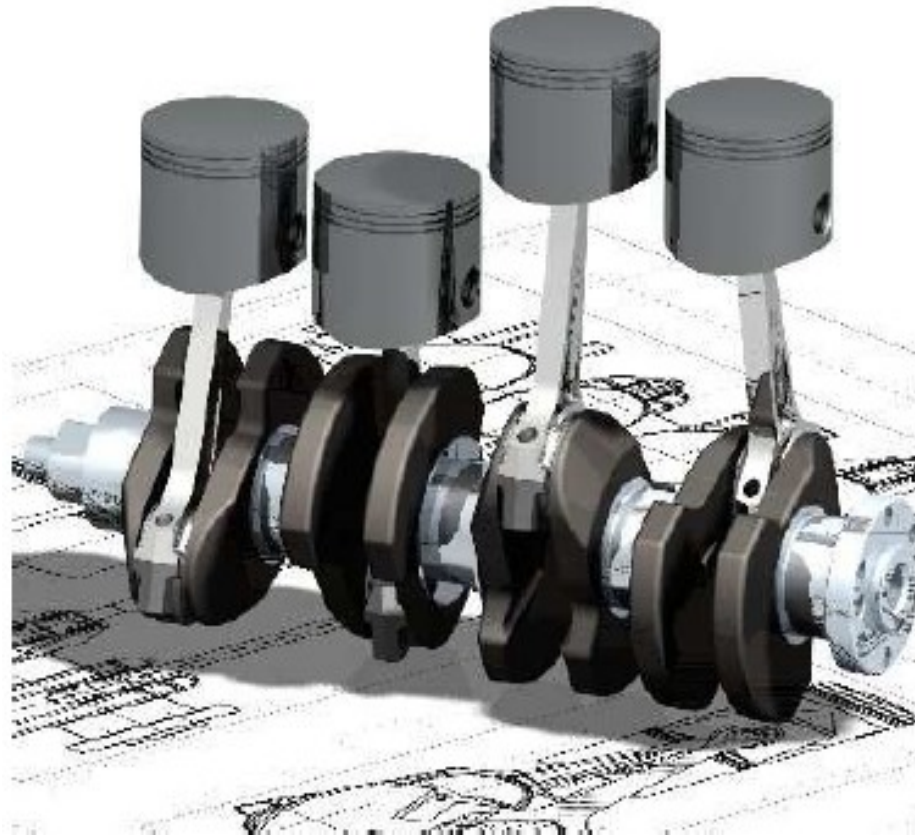
Application

❑ Computer-Aided Design – Architecture



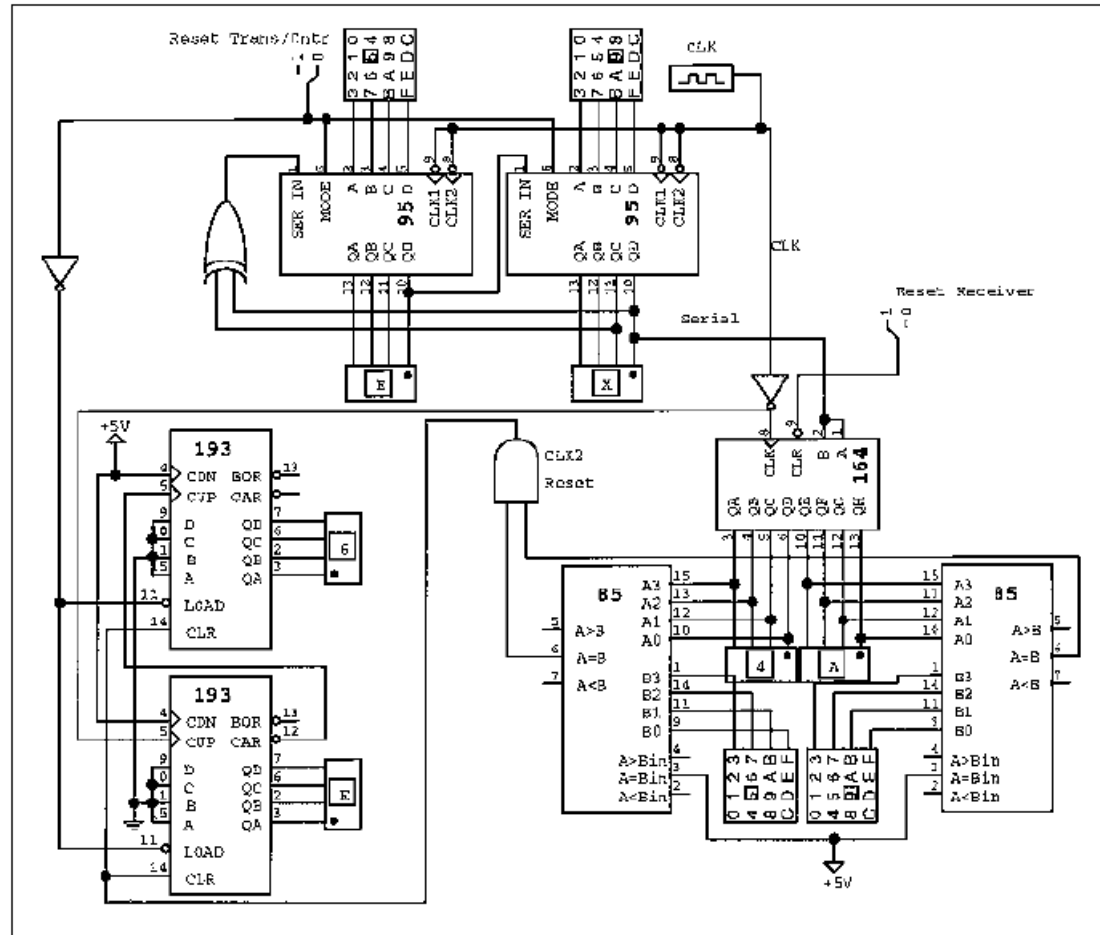
Application

- ❑ Computer-Aided Design
 - **Mechanical Engineering**



Application

- Computer-Aided Design
 - Digital Logic Design



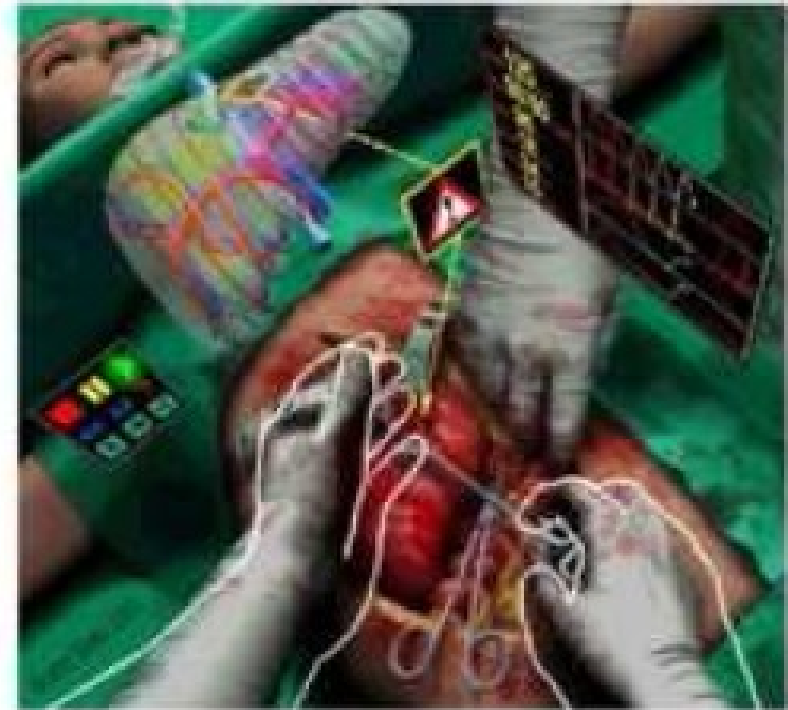
Application

- ❑ Simulation and Animation
 - **Flight simulators**



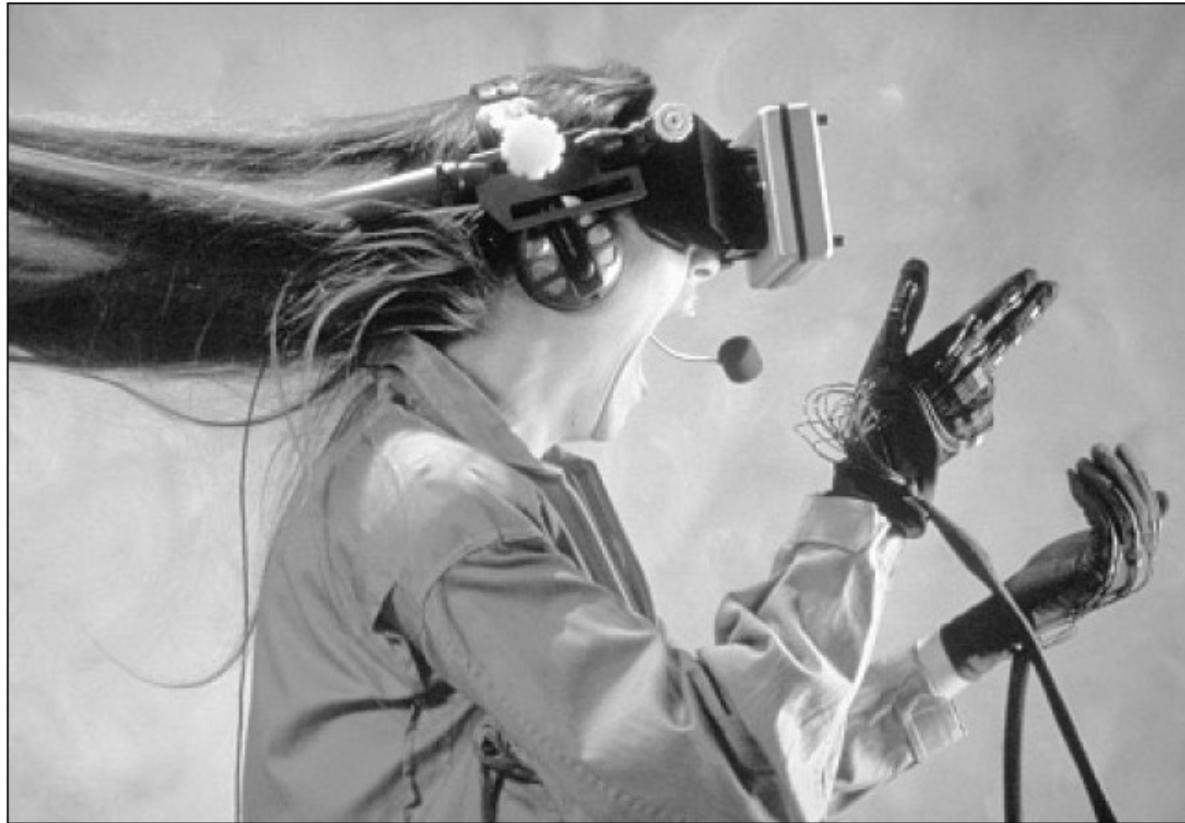
Application

- ❑ Simulation and Animation
 - **Surgical training**



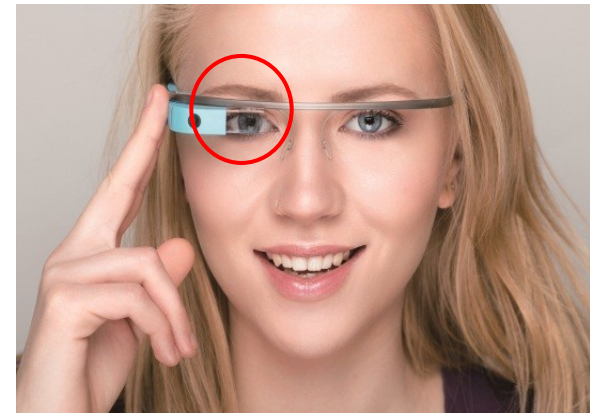
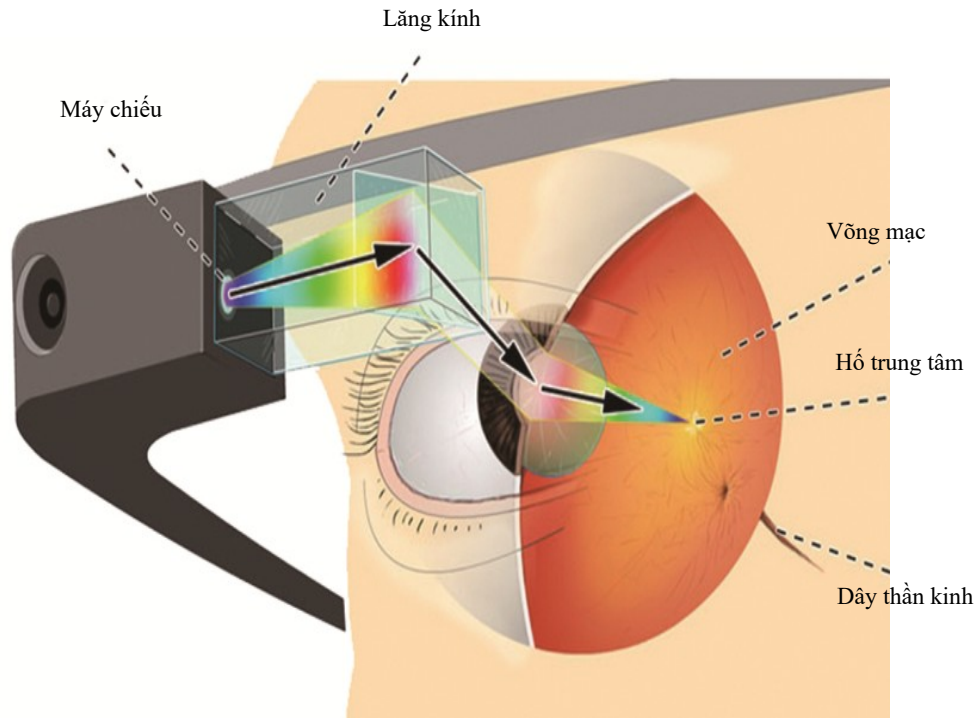
Application

- ❑ Simulation and Animation
 - Virtual Reality



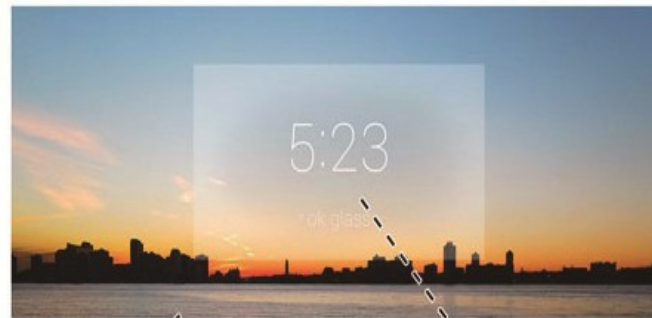
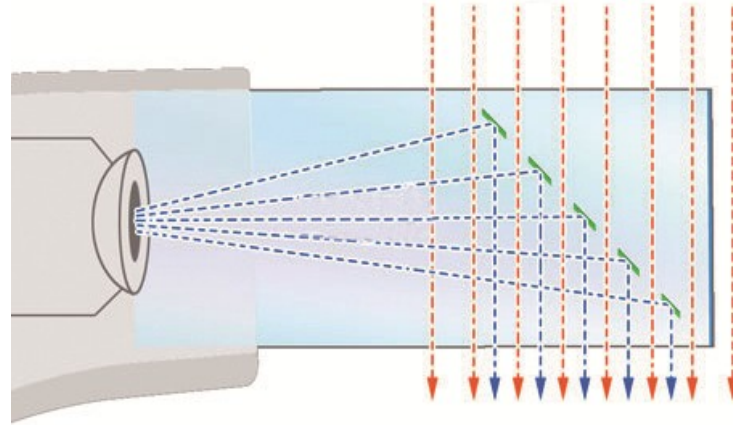
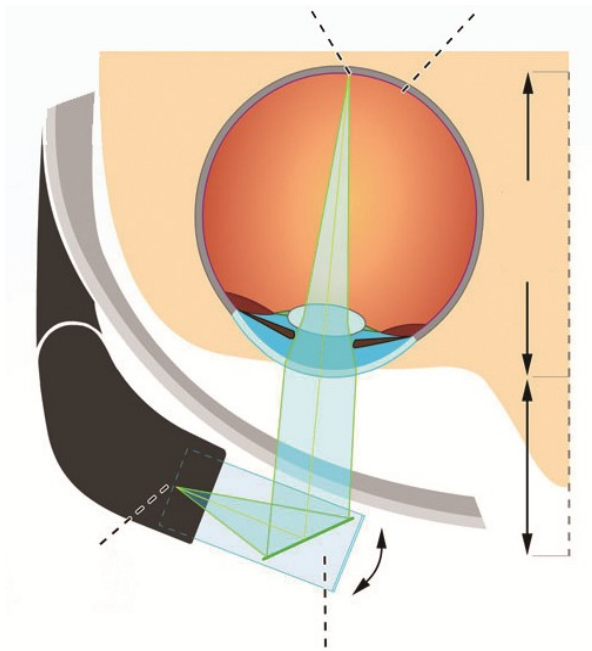
Application

❑ Simulation and Animation – **Augmented Reality**



Application

- ❑ Simulation and Animation
 - Augmented Reality



Cảnh vật thực tế

Ảnh ảo

Application

- ❑ Simulation and Animation
 - Entertainment



Application

❑ User Interface



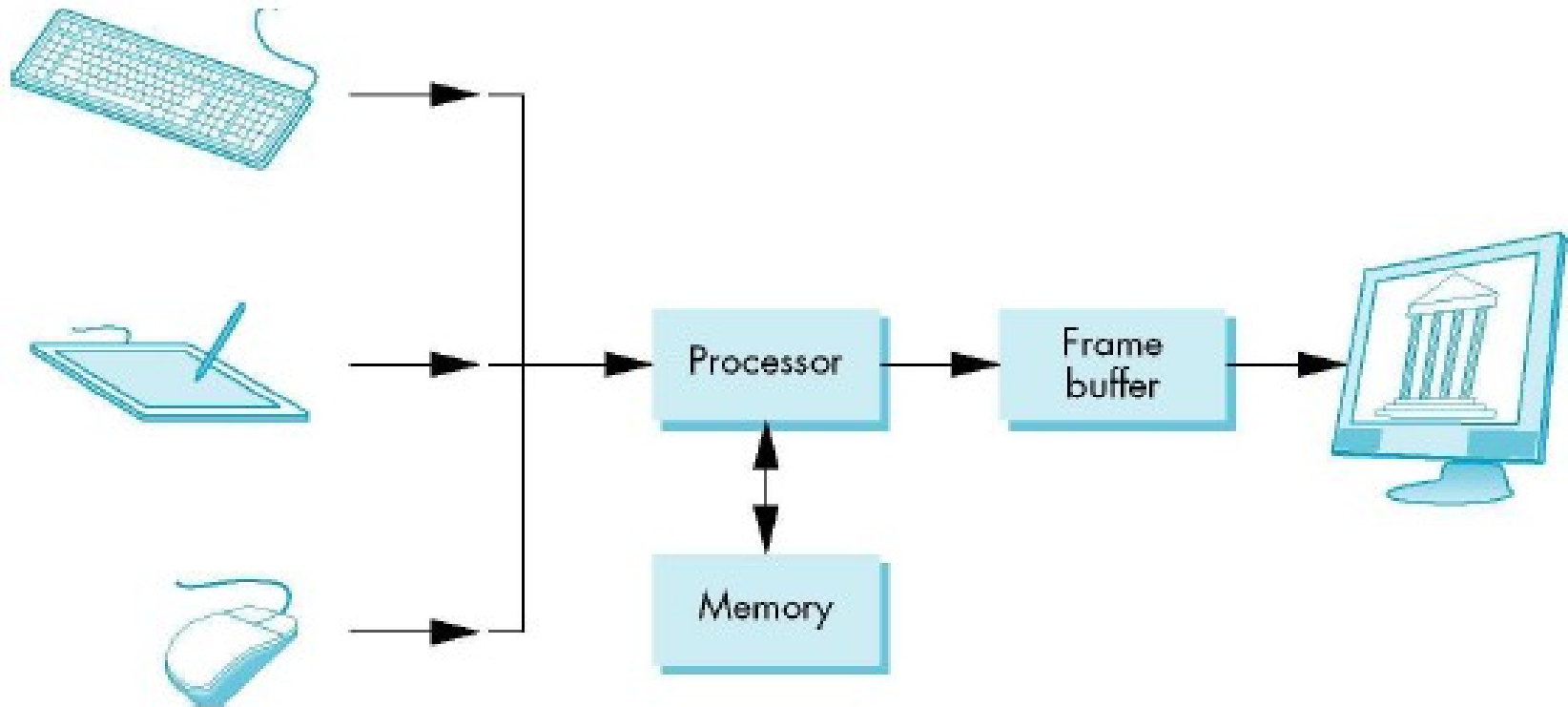
Computer Graphics Systems

❑ System's Overview

- Input devices
- Processor
- Memory
- Frame buffer
- Output devices

Computer Graphics Systems

❑ System's Overview

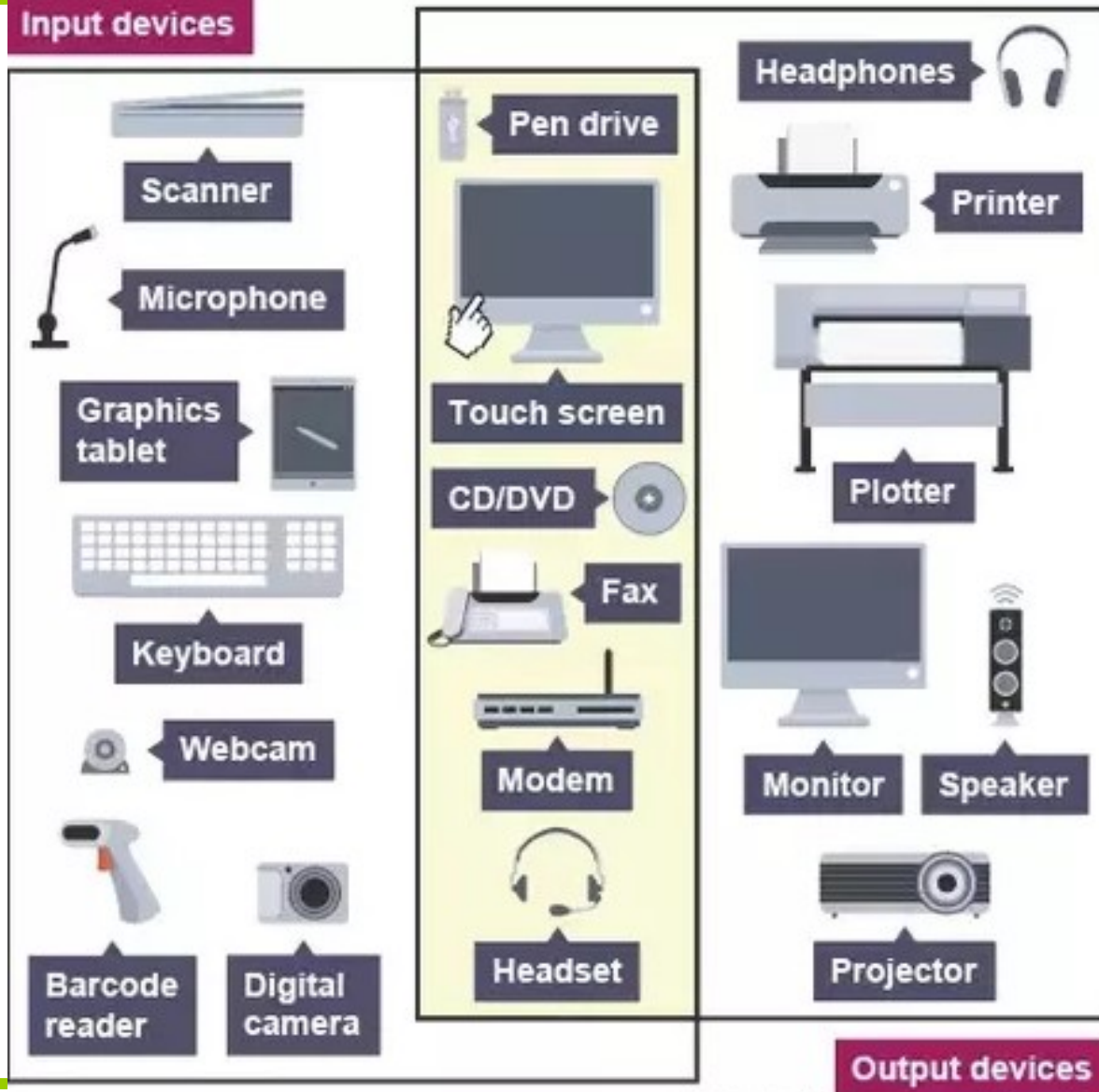


Computer Graphics Systems

□ Input devices

- **Keyboard, Button boxes, dials**
- **Mouse Devices: 2D and 3D**
- **Trackballs and Spaceballs**
- **Joysticks**
- **Data Gloves, CyberGloves**
- **Data tablet**
- **Image Scanner**
- **Touch Panels**
- **Light Pens**
- **3D Scanner**

Computer Graphics Systems



Computer Graphics Systems

❑ Frame buffer

– Location

- Inside the system memory
- Inside GPU, graphic card

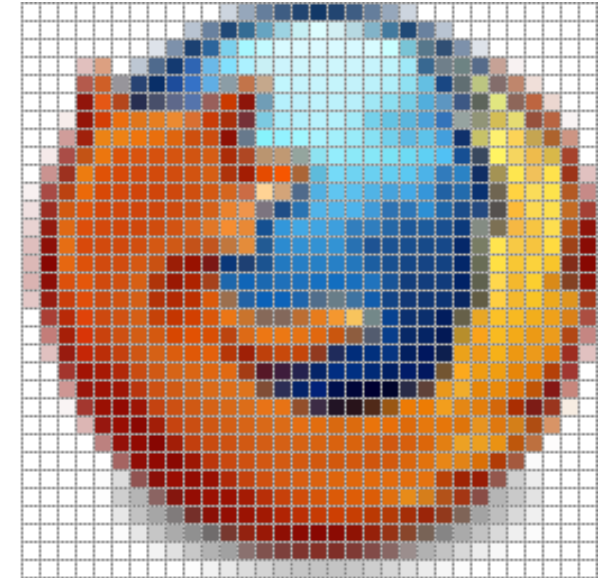
– **Store pixels of image to be shown on video display**

Computer Graphics Systems

□ Frame buffer

– Pixel

- **Is the smallest element of images**
- **Image = 2D Array of pixels**
- **Specification:**
 - Location: (X,Y)
 - Value: (Color)
 - Gray value
 - Color: [R,G,B]
 - Index to color



Computer Graphics Systems

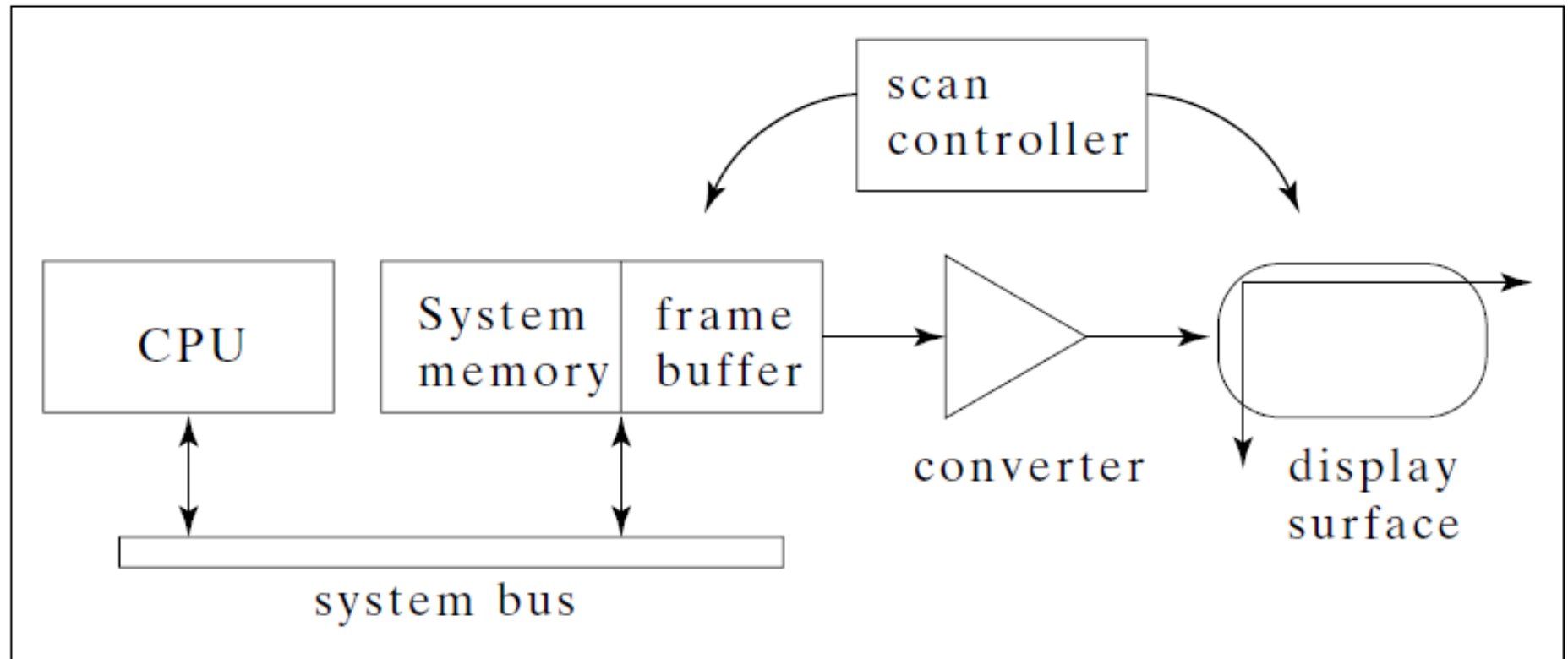
❑ Frame buffer

– **Specification:**

- Resolution: the number of pixels in the frame buffer
- Depth or Precision: the number of bits that are used for each pixel
 - 1 bit: black and white color
 - 8 bits: 28 (= 256) colors
 - 24 bits: full-color system or true-color system.

Computer Graphics Systems

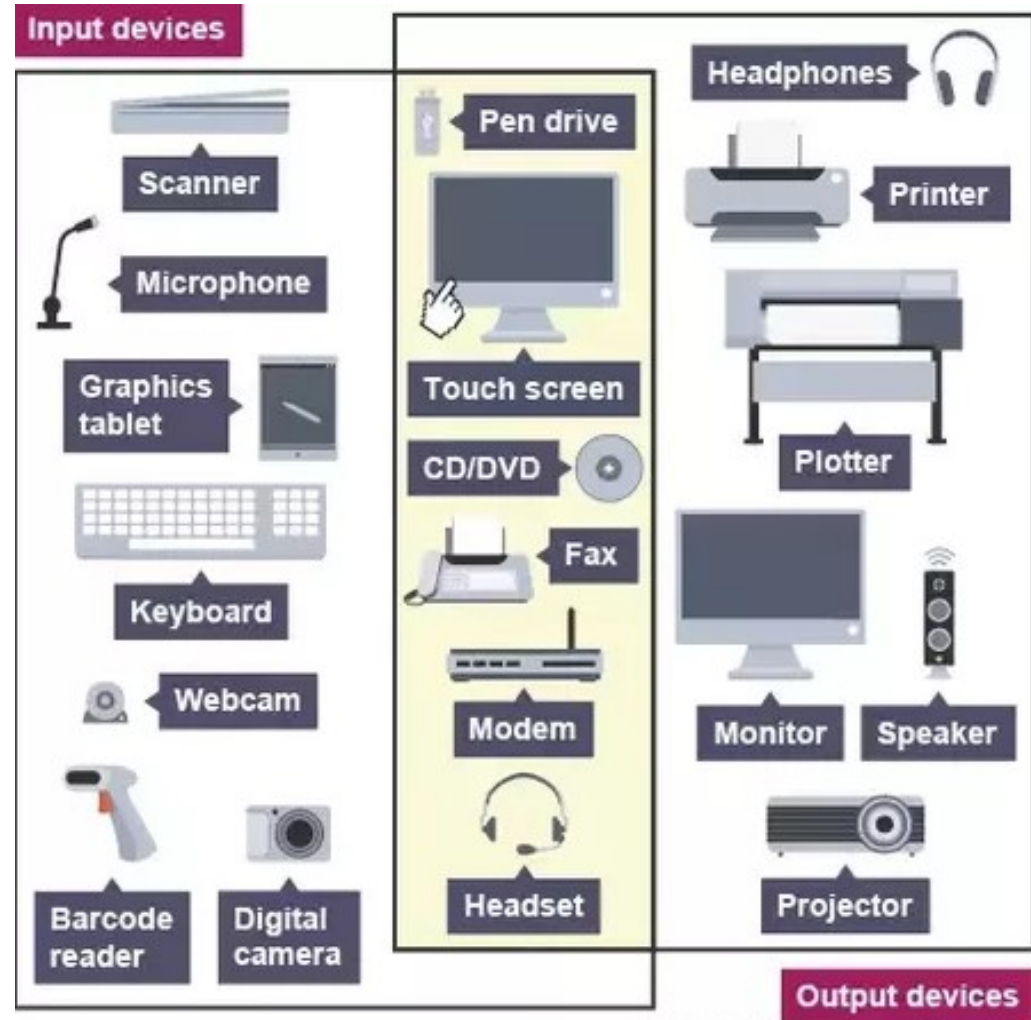
□ Frame buffer



Computer Graphics Systems

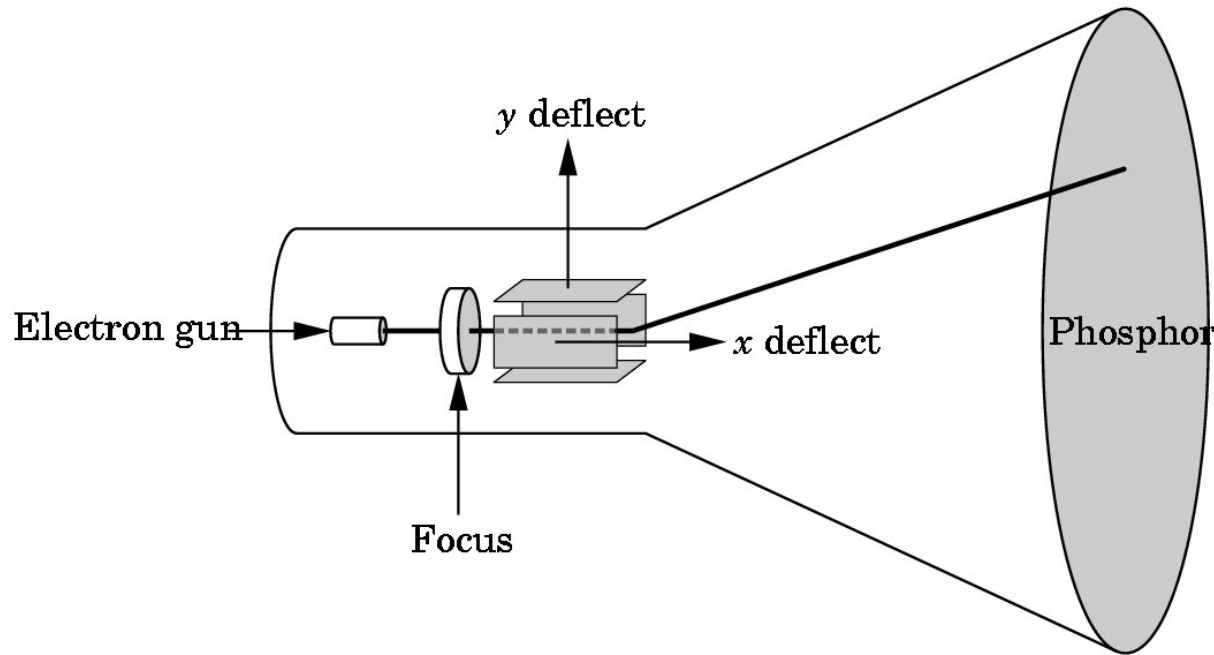
- ❑ Output devices
 - **Hard-copy devices**
 - **Printer**
 - **Film recorder**
 - **Video display/projector**
 - Cathode-Ray Tube (CRT)
 - Flat-panel display.

Computer Graphics Systems



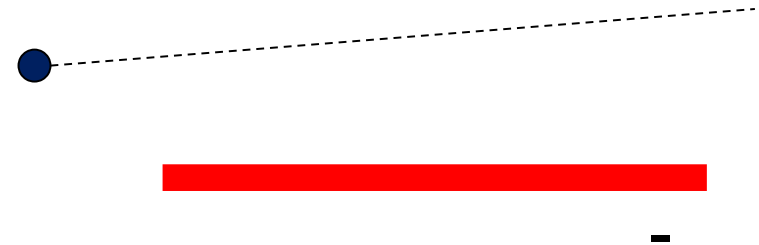
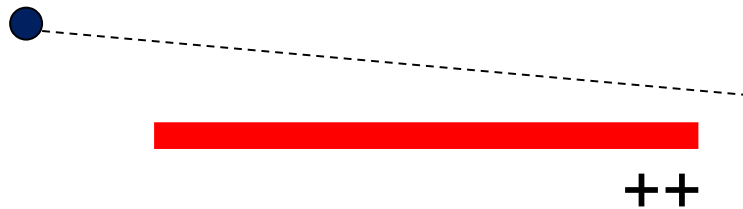
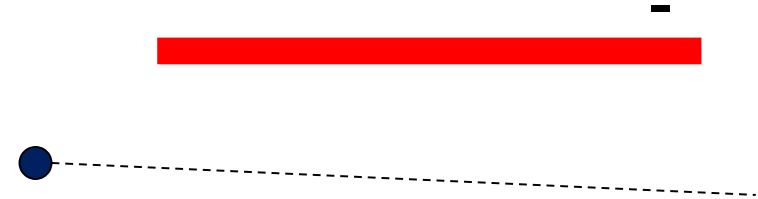
Computer Graphics Systems

- ❑ Output devices
 - CRT (*cathode ray tube*)



Computer Graphics Systems

□ Output Devices



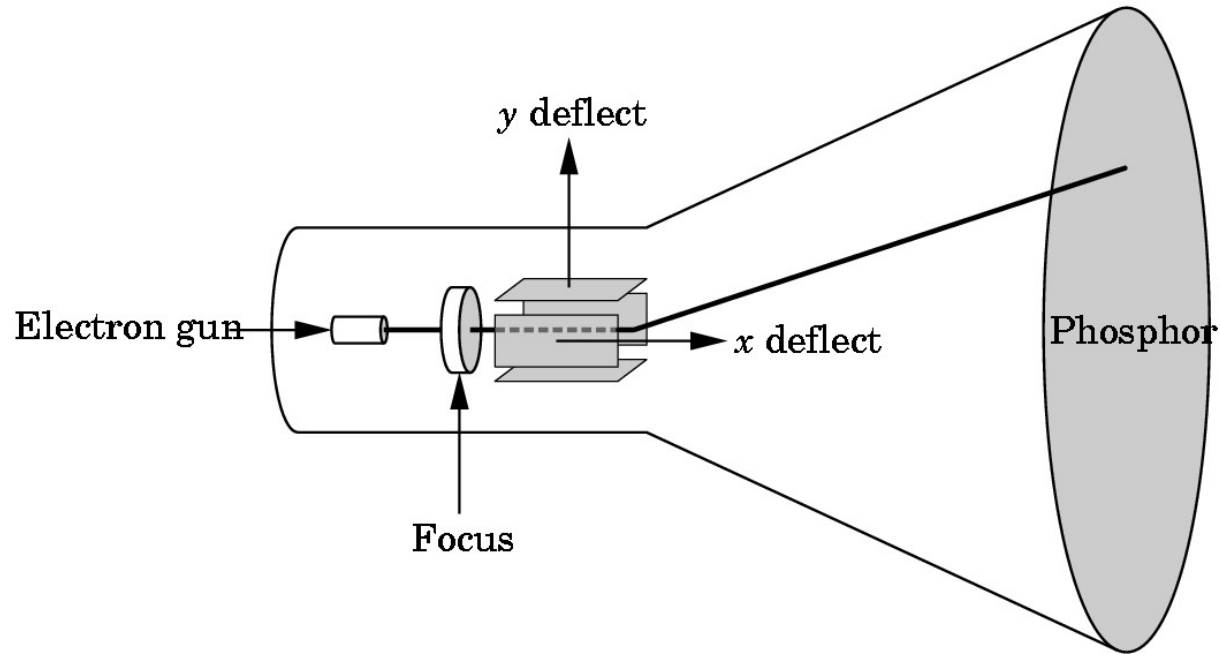
Computer Graphics Systems

❑ Output Devices



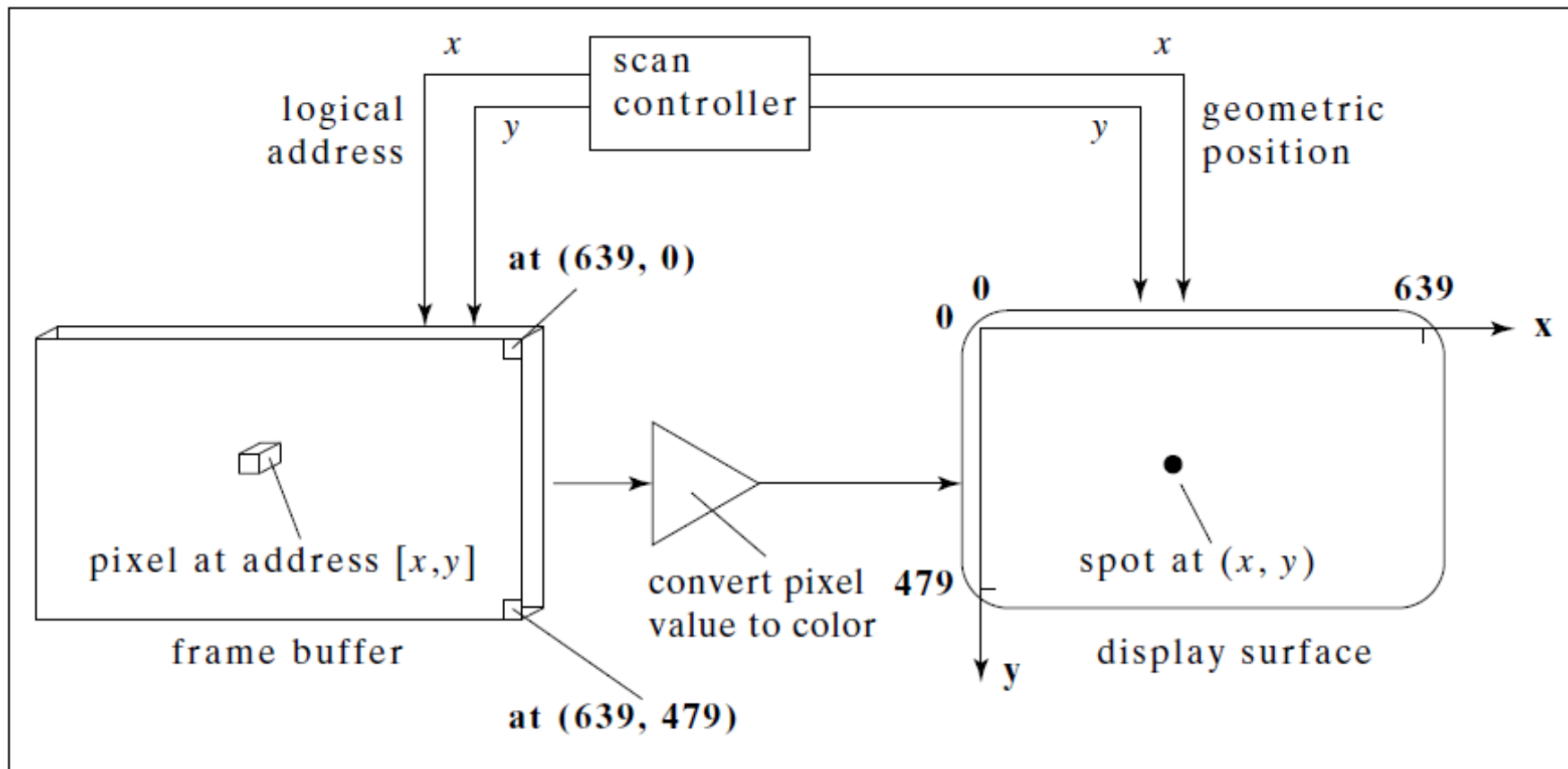
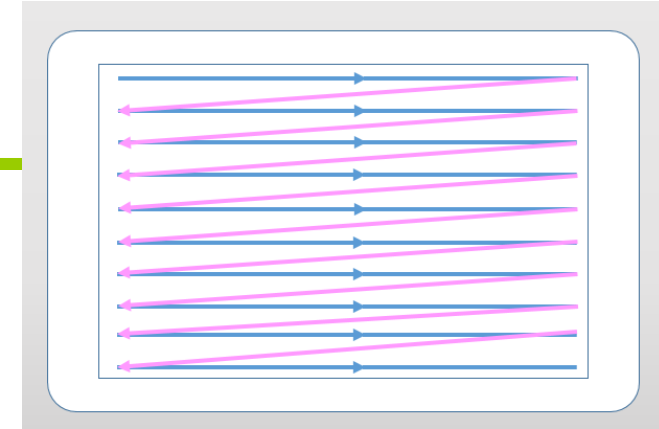
Computer Graphics Systems

□ Output Devices



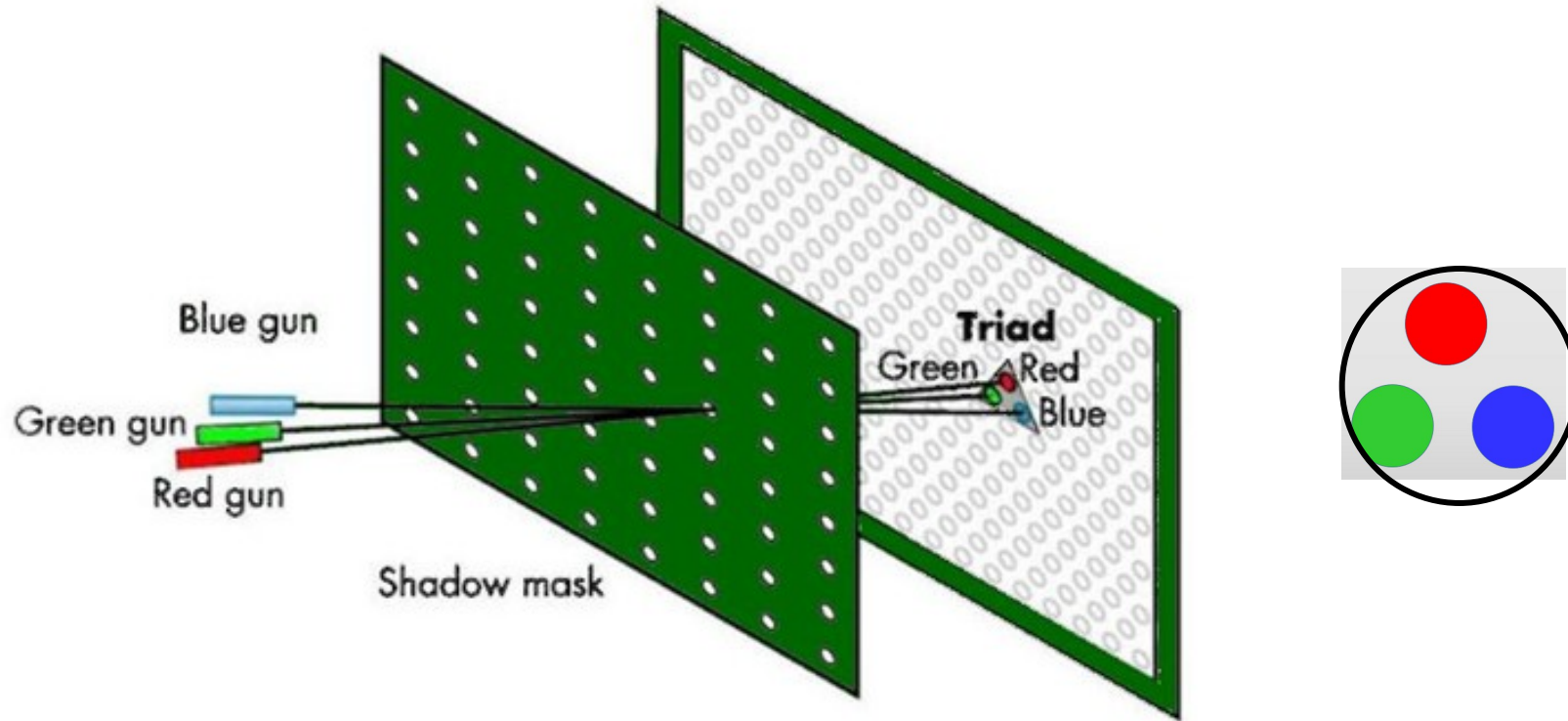
Computer Graphics Systems

□ Output devices



Computer Graphics Systems

□ Output Devices – Color Monitor

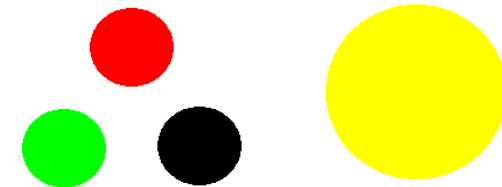


Computer Graphics Systems

□ Output Devices – Color Monitor



RGB =(0, 0, 0)



RGB =(255, 255, 0)



RGB =(255, 0, 0)



RGB =(0, 255, 0)



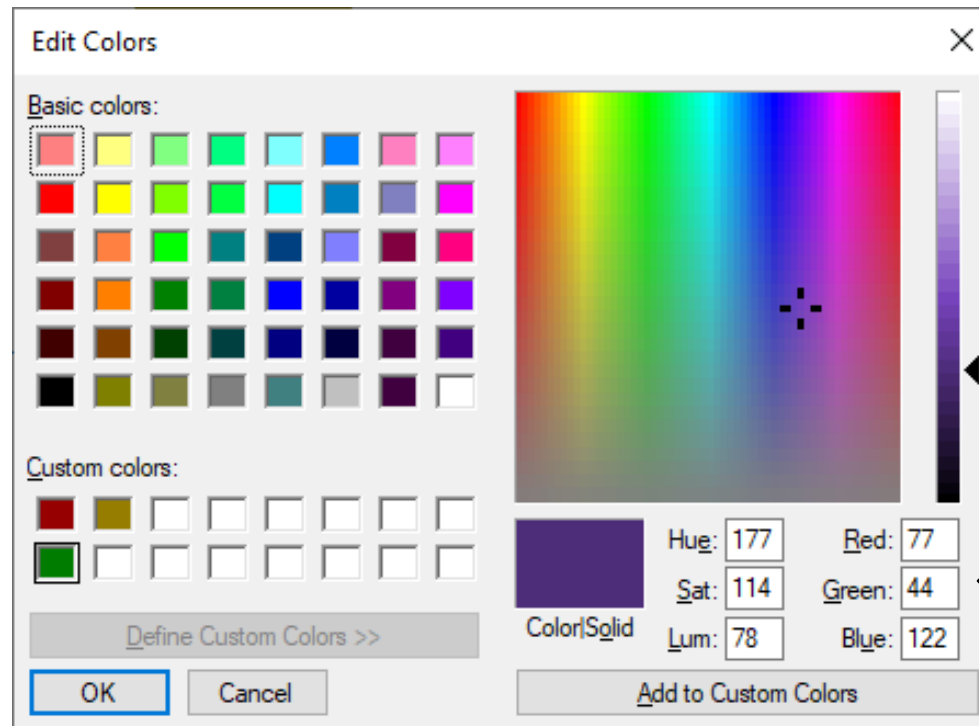
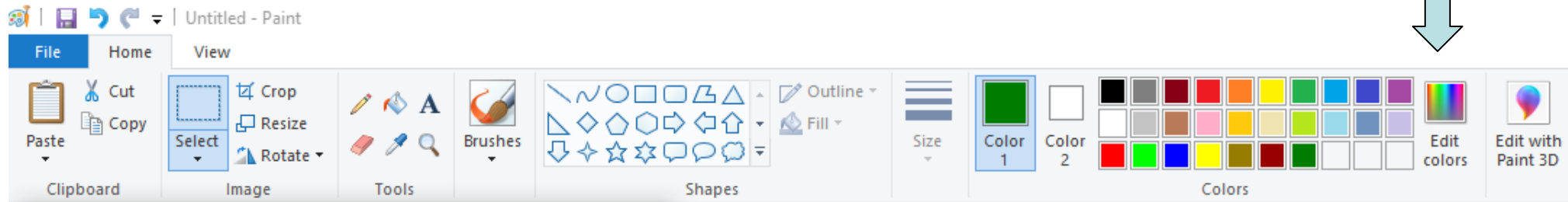
RGB =(150, 125, 0)



RGB =(0, 0, 255)

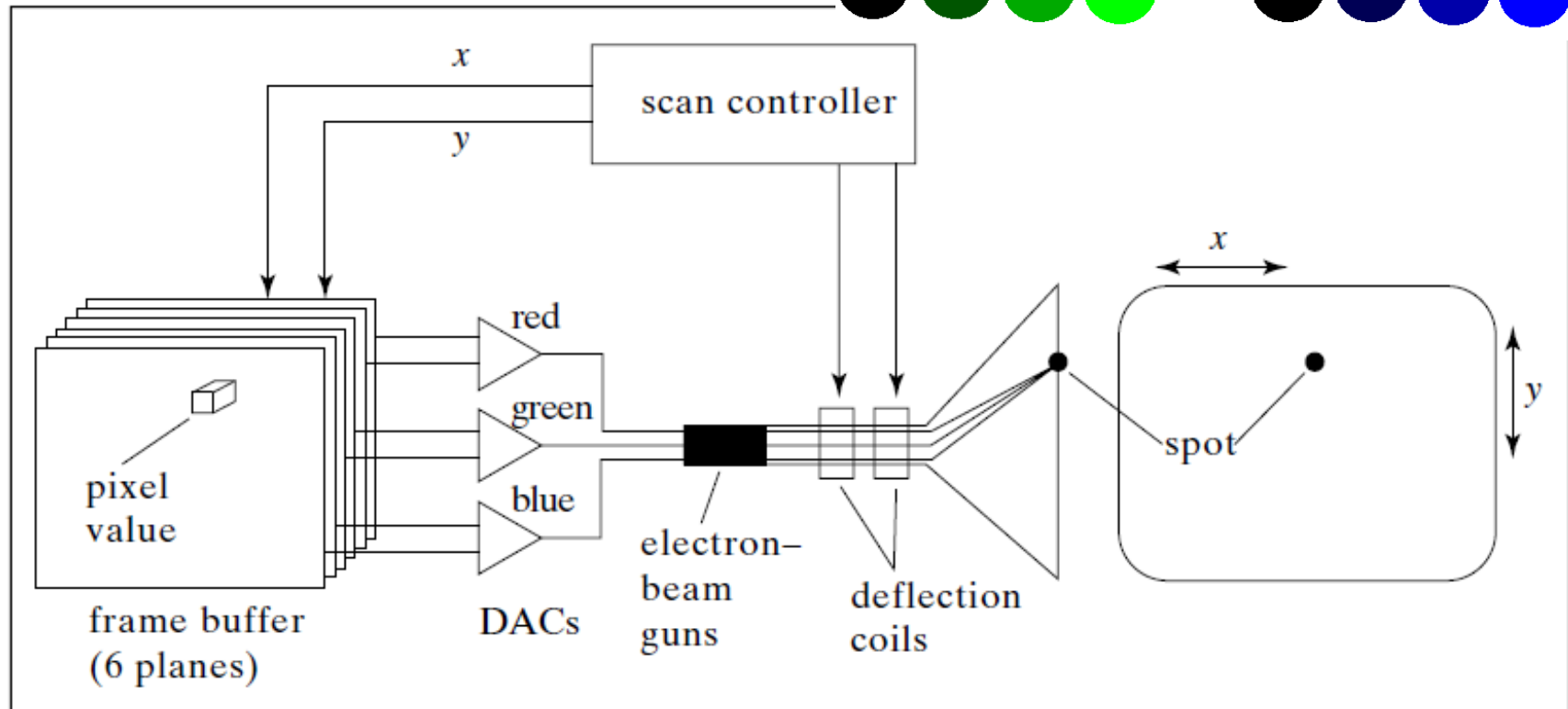
Computer Graphics Systems

Output Devices

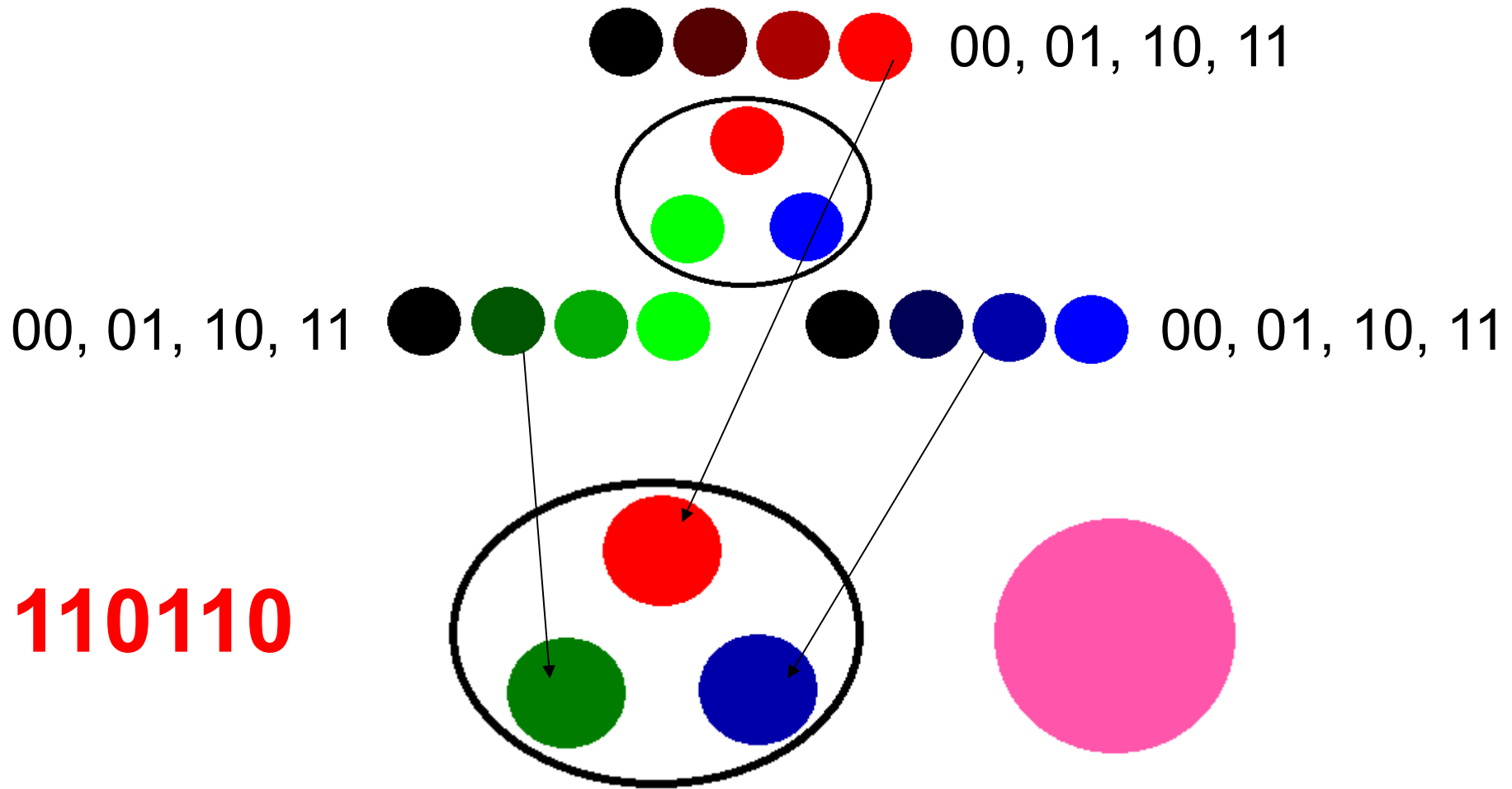


Computer Graphics Systems

- Output devices
 - Color CRT

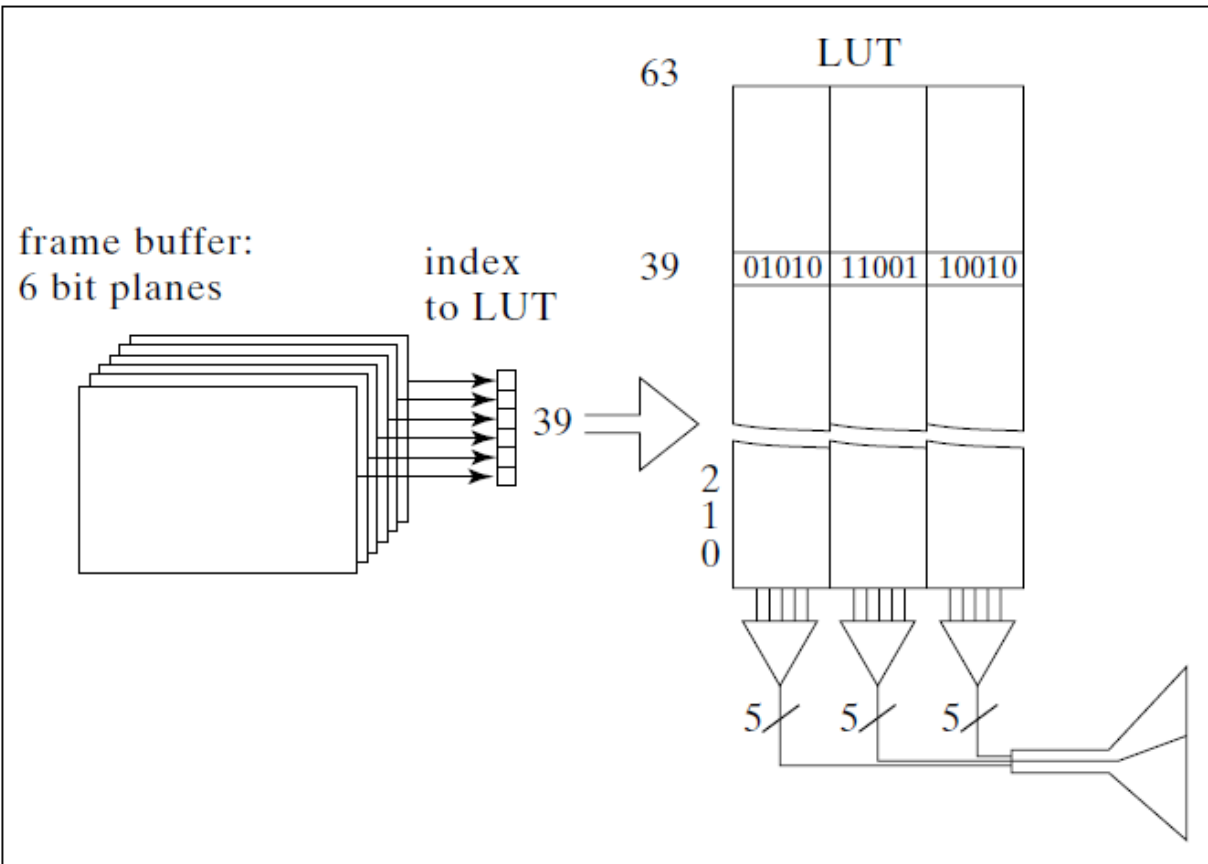


Computer Graphics Systems



Computer Graphics Systems

- ❑ Output devices
 - Indexed Color & Look up table



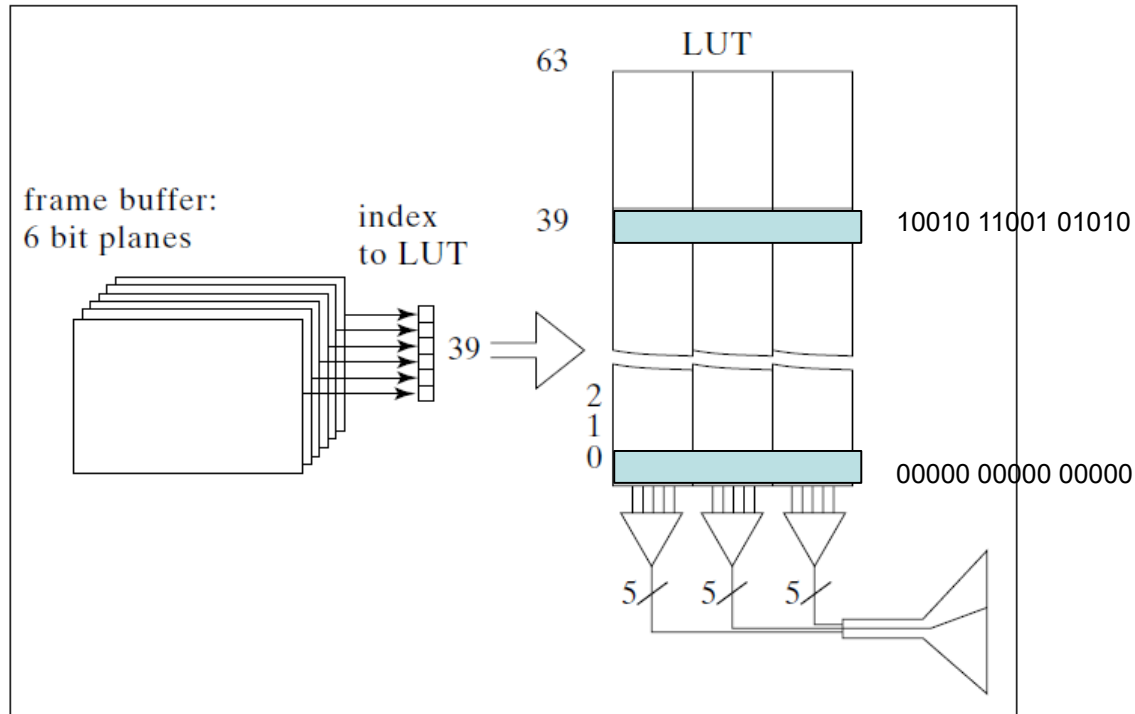
$$640 * 480 * 6 / 8 = 230 \text{ K}$$

$$640 * 480 * 15 / 8 = 230 \text{ K} * 2.5$$

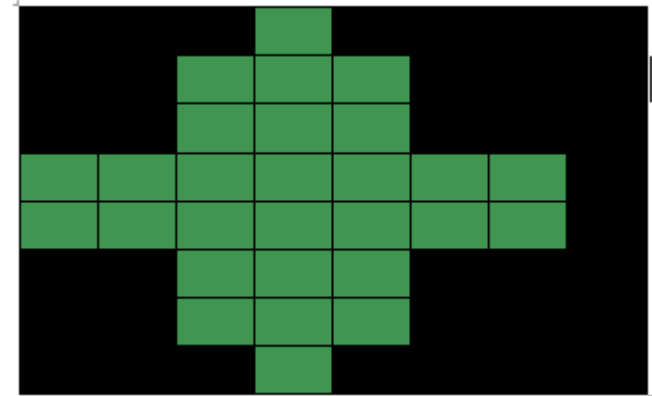
$$\text{LUT : } 64 * 15 / 8 = 120 \text{ bytes}$$

Computer Graphics Systems

□ Output devices



0	0	0	39	0	0	0	0
0	0	39	39	39	0	0	0
0	0	39	39	39	0	0	0
39	39	39	39	39	39	39	0
39	39	39	39	39	39	39	0
0	0	39	39	39	0	0	0
0	0	39	39	39	0	0	0
0	0	0	39	0	0	0	0

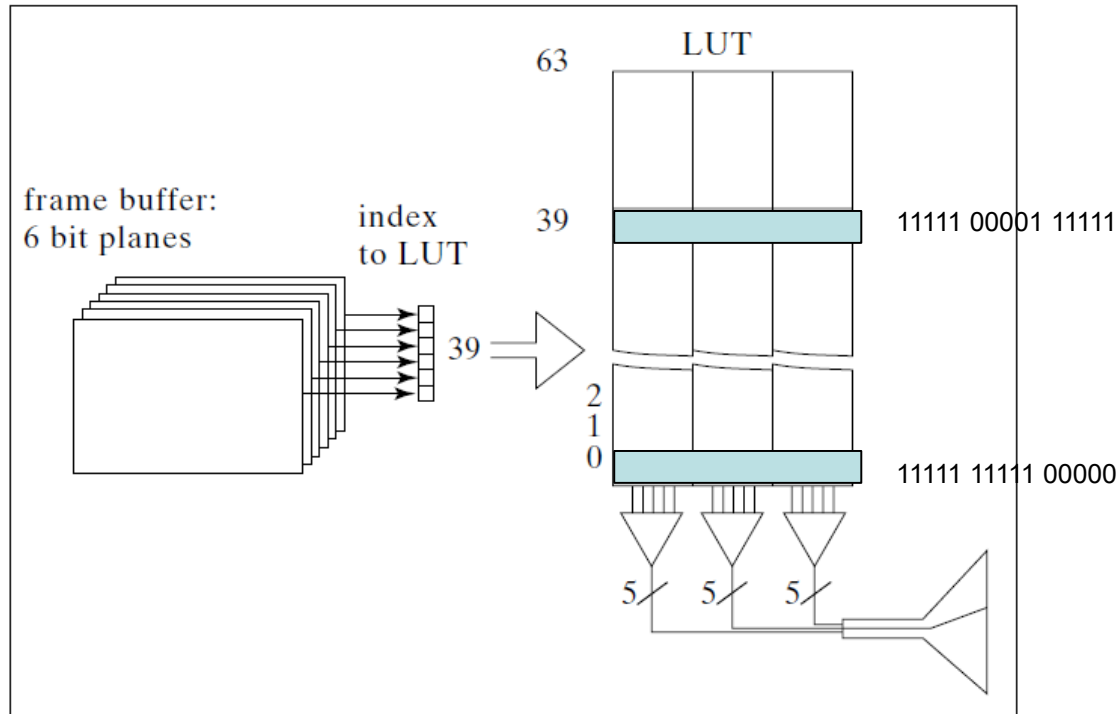


10010 = 18 11001 = 25 01010 = 10

$R = 18 \cdot 8 = 64$ $G = 25 \cdot 8 = 150$ $B = 10 \cdot 8 = 80$

Computer Graphics Systems

Output Devices

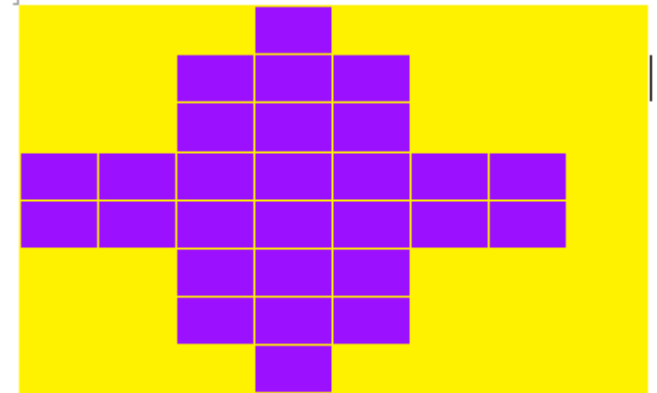


R, G, B

11111, 00010, 11111

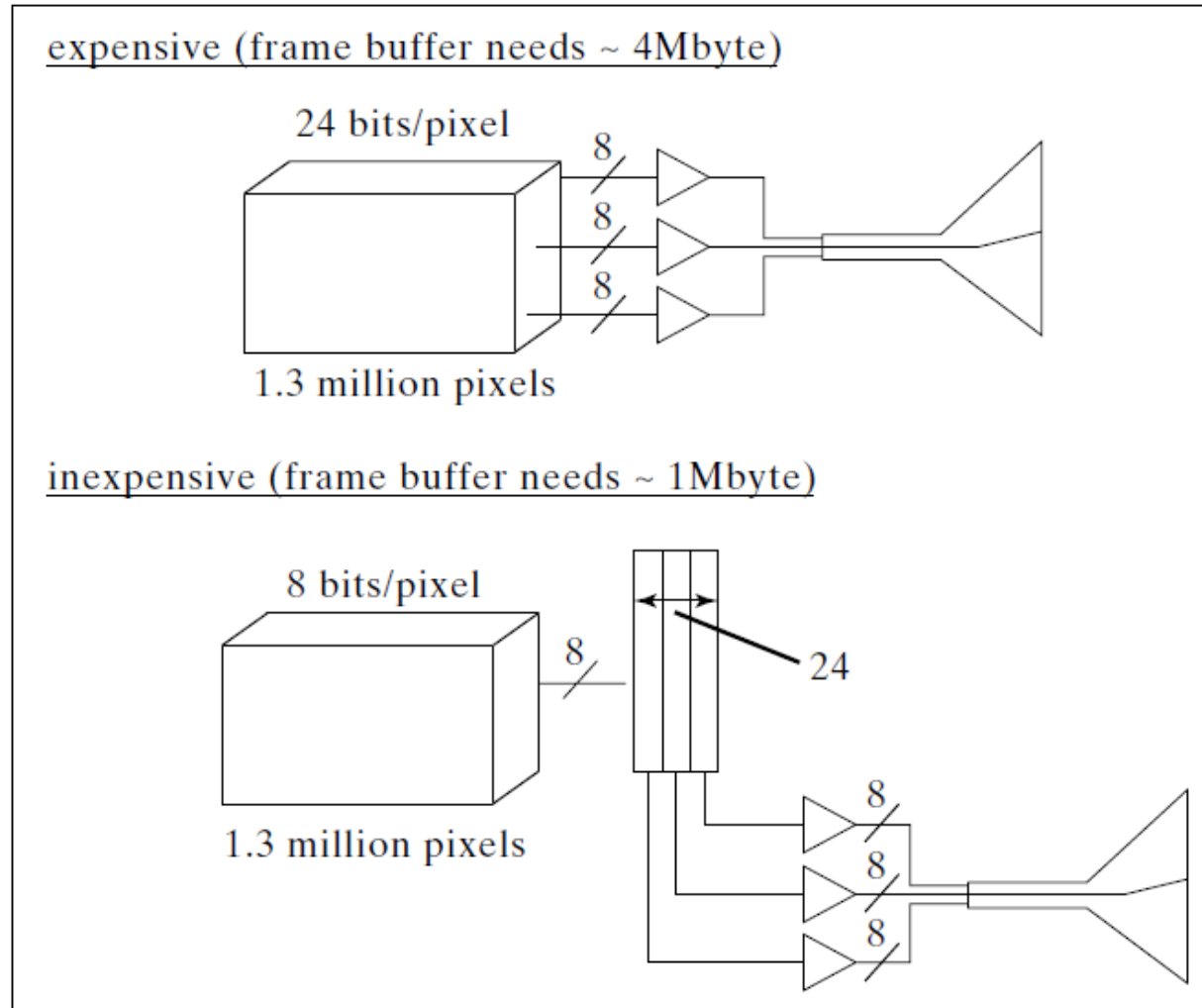
255 , 16, 255

0	0	0	39	0	0	0	0
0	0	39	39	39	0	0	0
0	0	39	39	39	0	0	0
39	39	39	39	39	39	39	0
39	39	39	39	39	39	39	0
0	0	39	39	39	0	0	0
0	0	39	39	39	0	0	0
0	0	0	39	0	0	0	0



Computer Graphics Systems

❑ Output devices



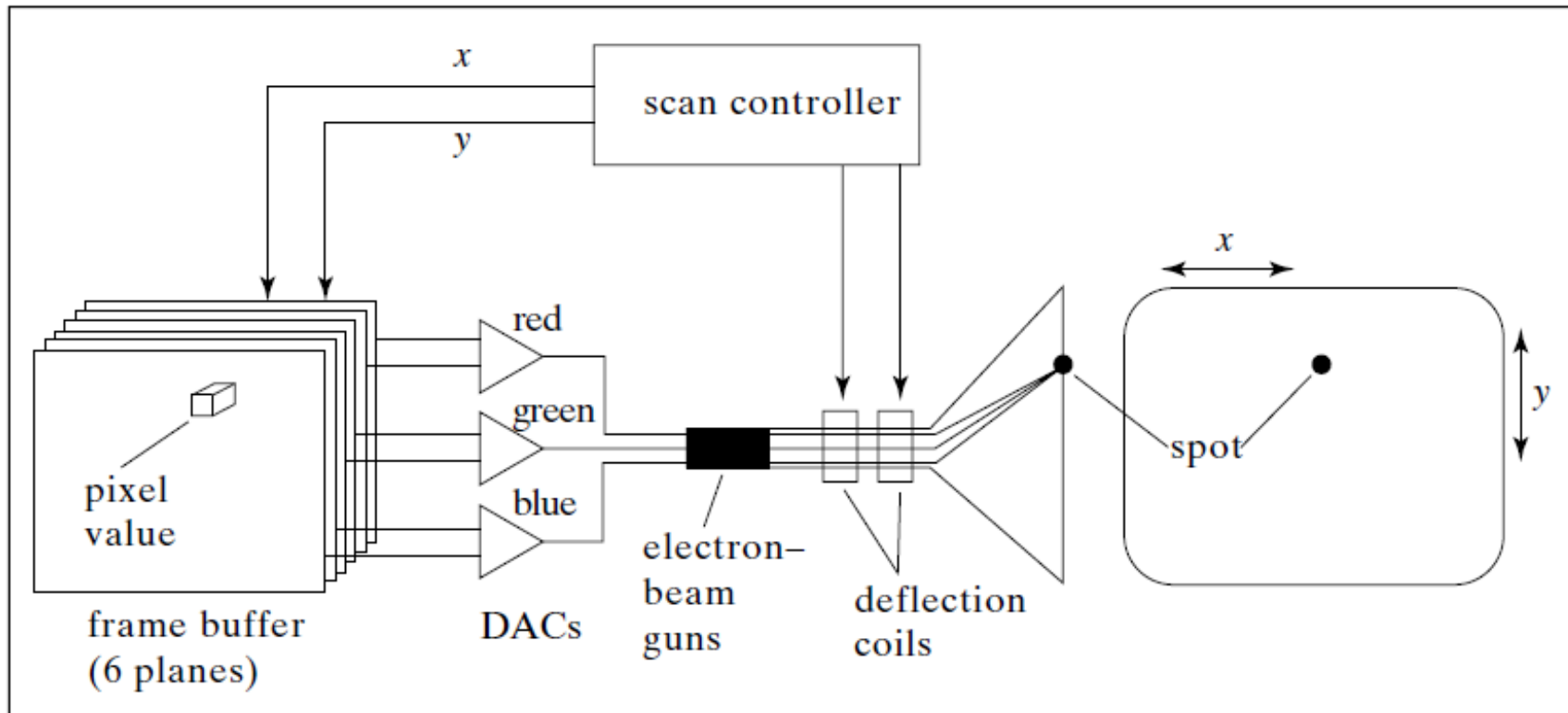
● RGB=(45,73,208)

● RGB=(45,74,208)

● RGB=(45,74,215)

Computer Graphics Systems

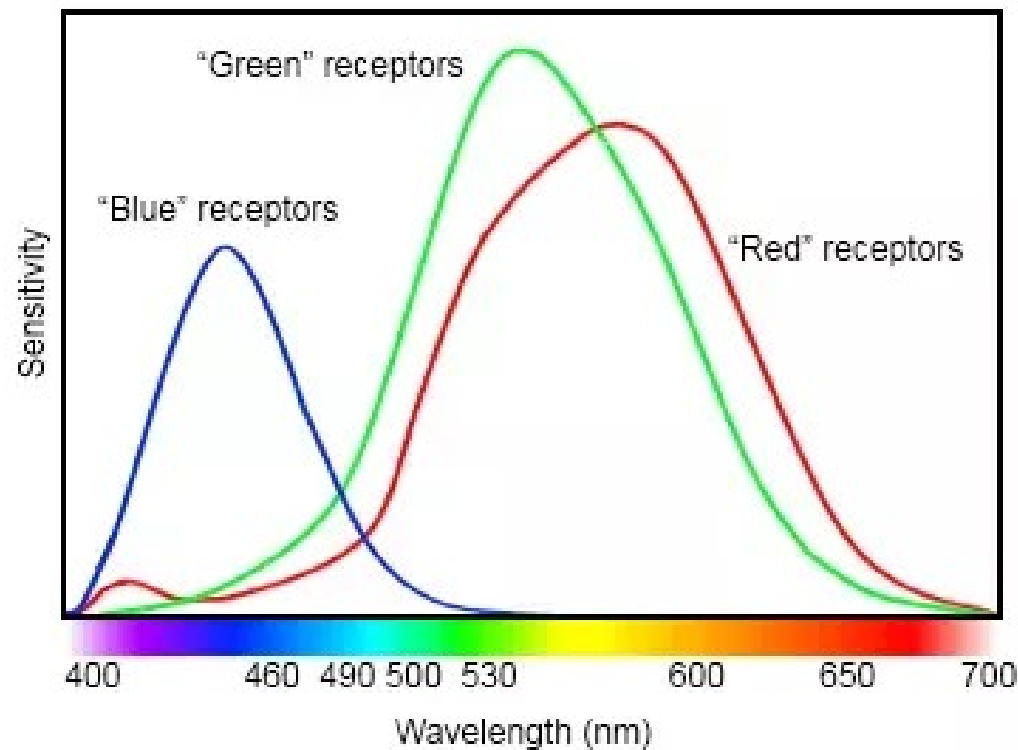
□ Output Devices



Computer Graphics Systems

□ Output Devices

Human color receptor relative sensitivity



Computer Graphics Systems

□ Output devices

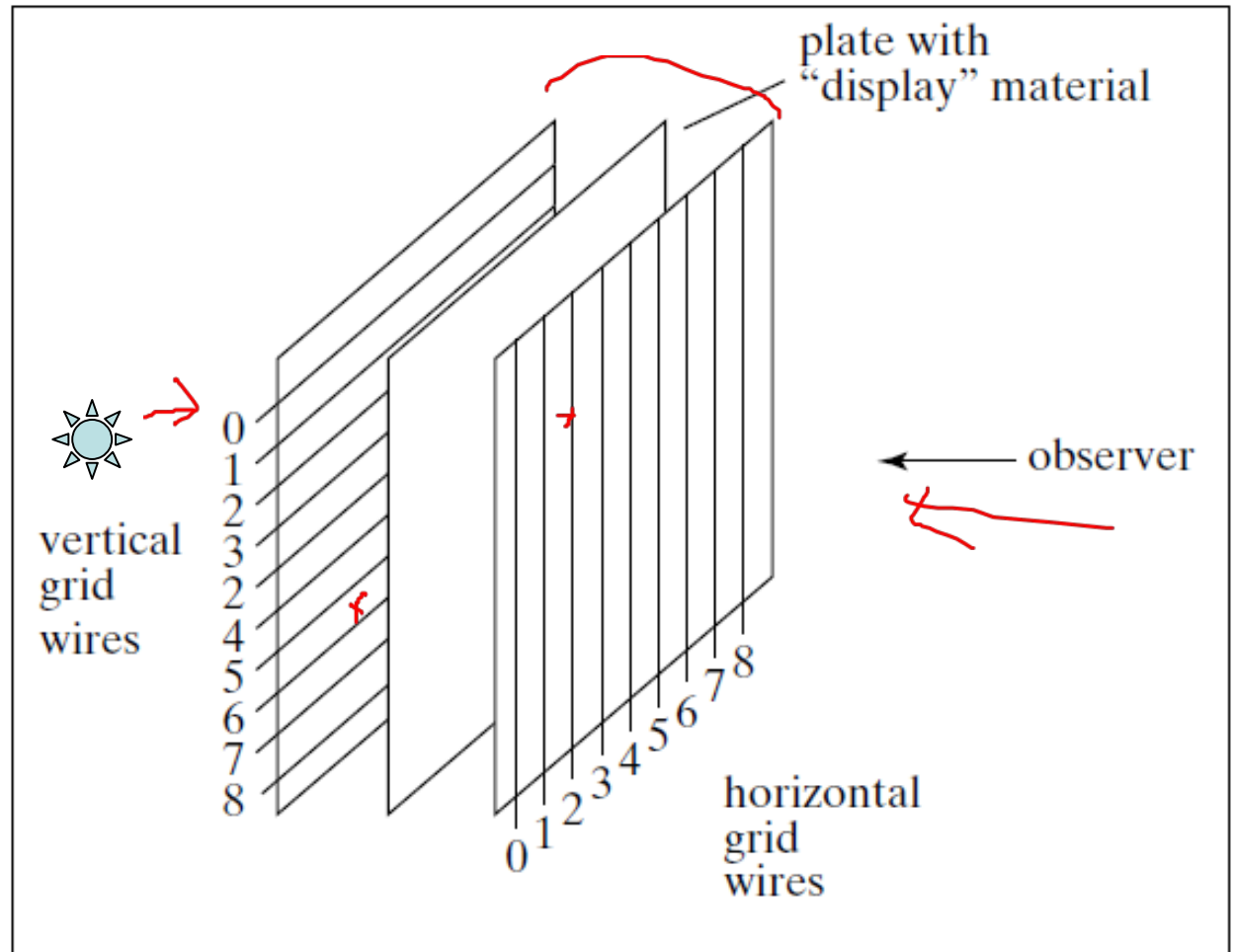


Image Formation

❑ Elements of Image Formation

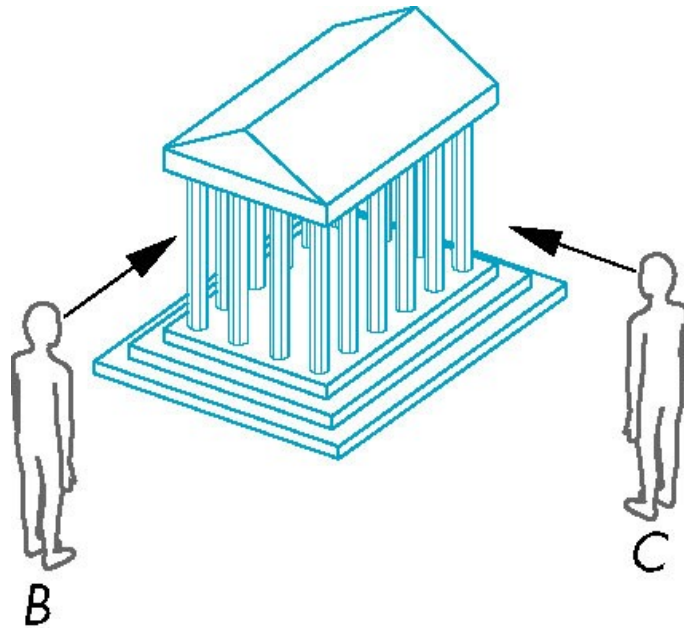
- Objects
- Viewers
- Lights (Materials)

❑ Advantages

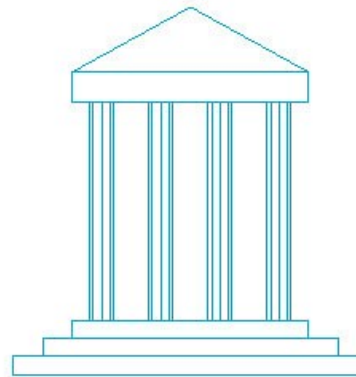
- Separation of objects, viewer, light sources
- Two-dimensional graphics is a special case of three-dimensional graphics
- Leads to simple software API
- Leads to fast hardware implementation

Image & Image Formation

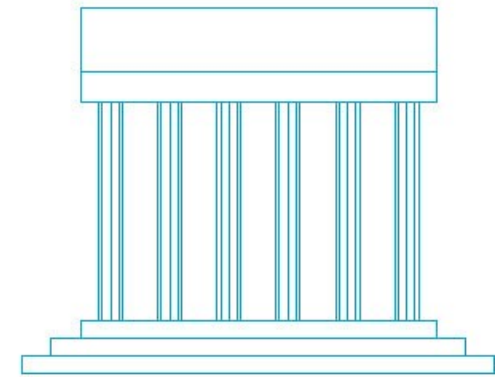
- Elements of Image Formation
 - Objects & viewers



(a)



(b)



(c)

Image & Image Formation

- ❑ Elements of Image Formation
 - Objects & viewers

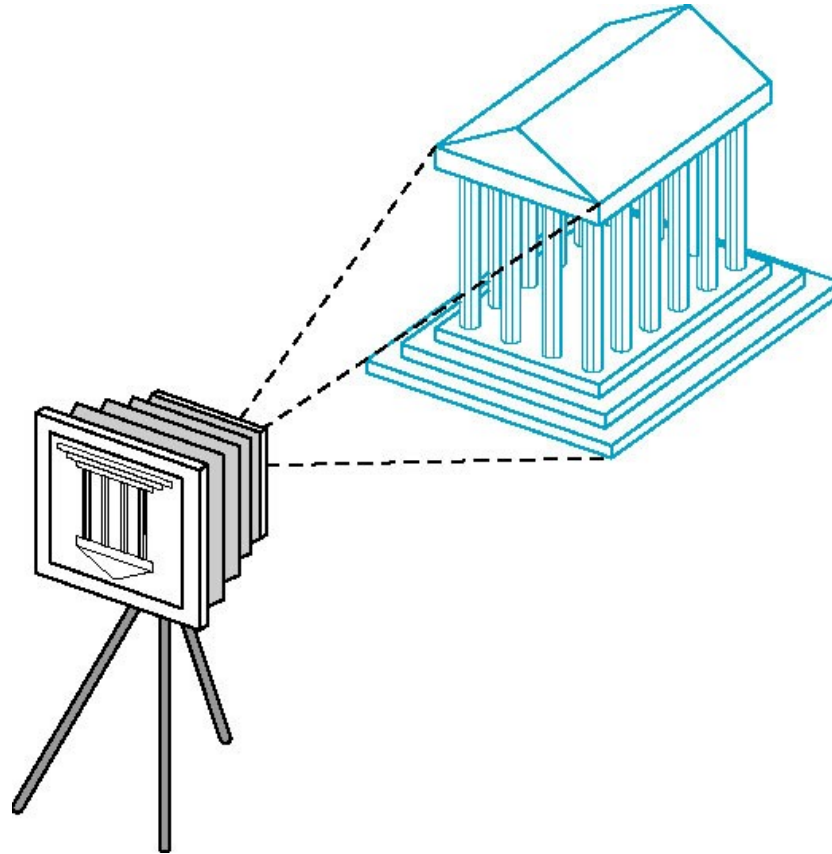
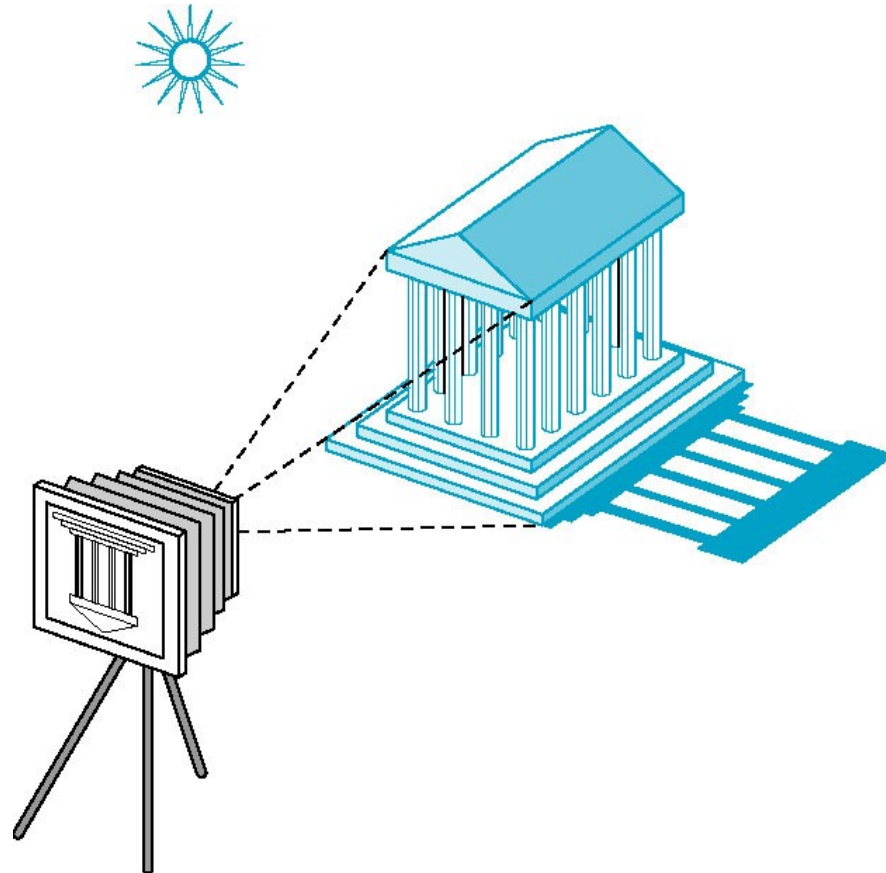


Image & Image Formation

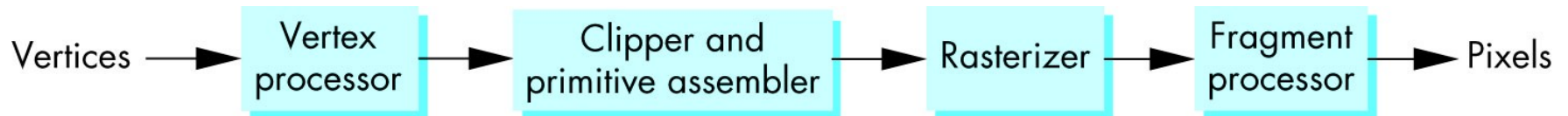
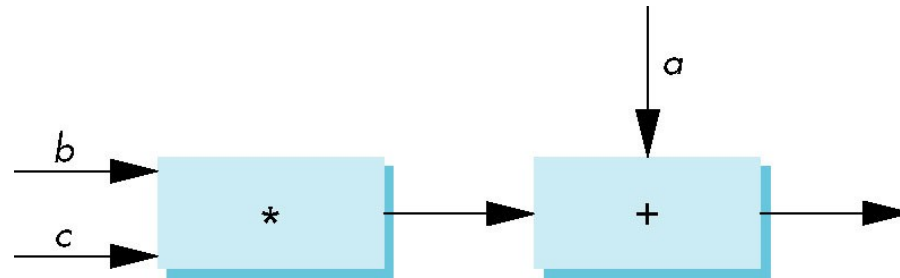
- ❑ Elements of Image Formation
 - Lights



Models and Architectures

□ Practical Approach : Pipeline

- Fast, simple
- All steps can be implemented in hardware on the graphics card



API Contents

- ❑ Functions that specify what we need to form an image
 - **Objects**
 - **Viewer**
 - **Light Source(s)**
 - **Materials**
- ❑ Other information
 - **Input from devices such as mouse and keyboard**
 - **Capabilities of system**

API Contents

❑ Object Specification

- Most APIs support a limited set of primitives including
 - **Points (0D object)**
 - **Line segments (1D objects)**
 - **Polygons (2D objects)**
 - **Some curves and surfaces**
- All are defined through locations in space or *vertices*

API Contents

❑ Object Specification

The diagram illustrates the OpenGL API for specifying a triangle. It shows a sequence of function calls: `glBegin(GL_POLYGON)`, three `glVertex3f` calls with coordinates (0.0, 0.0, 0.0), (0.0, 1.0, 0.0), and (0.0, 0.0, 1.0), and finally `glEnd()`. Annotations with arrows point to specific parts of the code: 'type of object' points to `GL_POLYGON`, 'location of vertex' points to the first `glVertex3f` call, and 'end of object definition' points to `glEnd()`.

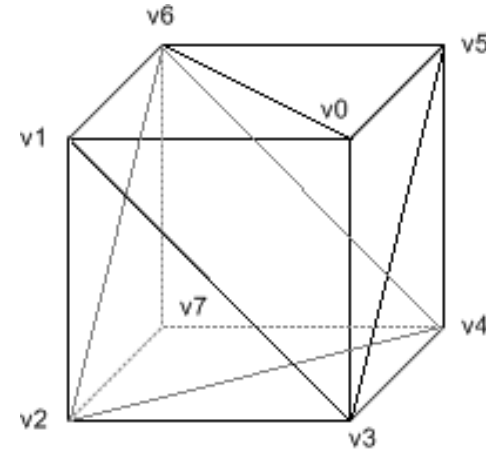
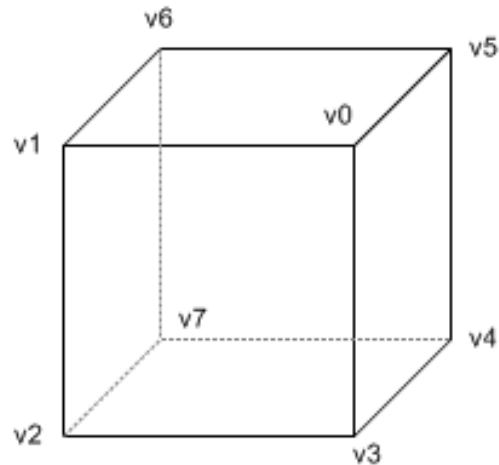
```
glBegin(GL_POLYGON)
  glVertex3f(0.0, 0.0, 0.0);
  glVertex3f(0.0, 1.0, 0.0);
  glVertex3f(0.0, 0.0, 1.0);
glEnd();
```

Annotations:

- type of object (points to `GL_POLYGON`)
- location of vertex (points to the first `glVertex3f` call)
- end of object definition (points to `glEnd()`)

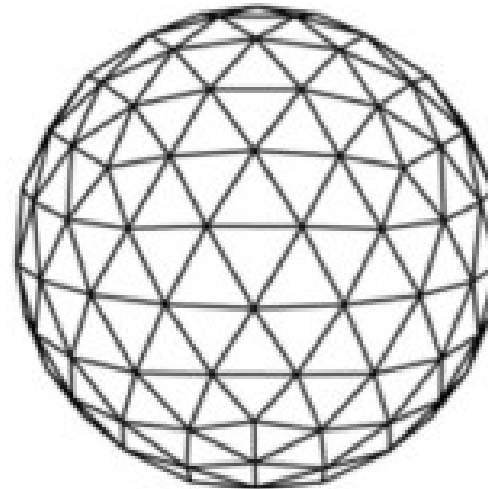
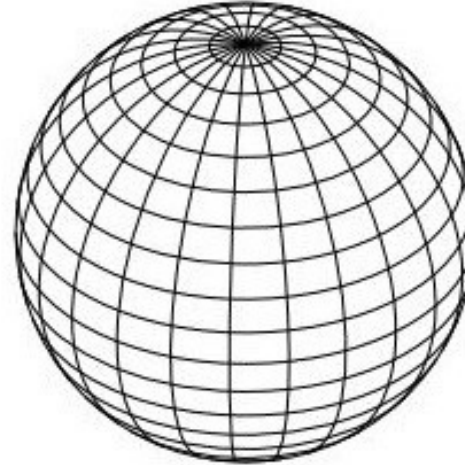
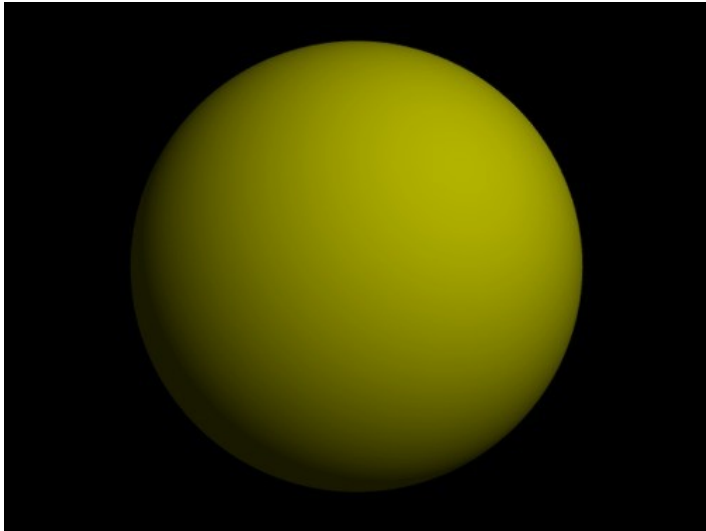
API Contents

❑ Object Specification



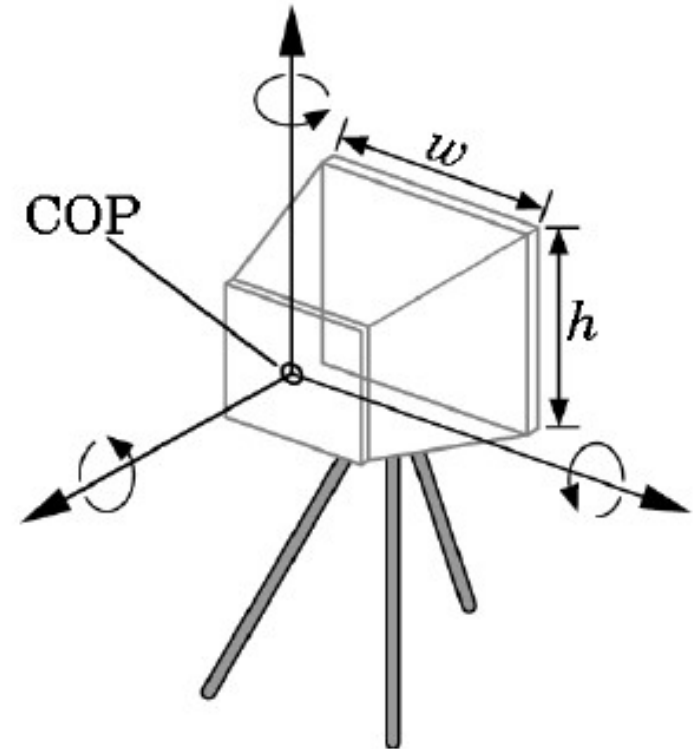
API Contents

❑ Object Specification



API Contents

- ❑ Camera Specification
 - Six degrees of freedom
 - **Position of center of lens**
 - **Orientation**
 - **Lens**
 - **Film size**
 - **Orientation of film plane**



API Contents

❑ Lights and Materials

– Types of lights

- **Point sources vs distributed sources**
- **Spot lights**
- **Near and far sources**
- **Color properties**

– Material properties

- **Absorption: color properties**
- **Scattering**
 - Diffuse
 - Specular

Further Reading

- ❑ **“Interactive Computer Graphics: A Topdown Approach Using OpenGL”, *Edward Angel***
 - Chapter 1: Graphics Systems And Models
- ❑ **“Đồ họa máy tính trong không gian hai chiều”, Trần Giang Sơn**
 - Chương 1: Giới thiệu đồ họa máy tính